

AUTOMATED TRANSCRIPT SYSTEM



Muhammad Abdul Rafay - 12296

Muhammad Mudassar Hassan - 23988

Muhammad Altahash Butt - 12366

Supervised By

Ms. Wajiha Mughal

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ABSTRACT

The Automated Transcript System, developed specifically for the National University of Modern Languages (NUML), revolutionizes the management of academic records for BS programs. Utilizing cutting-edge .NET technologies, including ASP.NET MVC Core, C#, and MS SQL Server Management Studio for robust data storage, the system integrates seamlessly with web technologies such as HTML, CSS, Bootstrap, and JavaScript within the Visual Studio 2022 environment.

This web-based application provides a user-centric experience, offering students an intuitive platform for real-time access to their result cards, unofficial transcripts, and course schedules. Coordinators benefit from streamlined record management, while exam branch staff can efficiently generate official transcripts, facilitating a more effective evaluation process.

Employing a waterfall iterative approach, the system ensures structured and systematic development. For testing, we apply black box testing techniques, leveraging Selenium to validate functionality like Un-Official and Official Transcripts, Results Management, Complete and Incomplete DMC's and ensure performance meets user expectations.

The Automated Transcript System is not merely a tool but a significant advancement towards a more efficient, secure, and accessible academic management system. Its design emphasizes fault tolerance, scalability, and stringent data security, ensuring a seamless operation within NUML's academic framework and significantly enhancing the overall student experience.

CERTIFICATE

It is certified that project report titled “**AUTOMATED TRANSCRIPT SYSTEM**” submitted by **Muhammad Abdul Rafay, Muhammad Mudassar Hassan** and **Muhammad Altahash Butt** for the partial fulfillment of the requirement of “**Bachelor’s Degree in Software Engineering**” is approved.

COMMITTEE

Dr. Muhammad Noman Malik
Dean Engineering & Computing

Signature: _____

Dr. Sumaira Nazir
HoD Engineering

Signature: _____

Mr. Naveed Ahmed
Head Project Committee

Signature: _____

Ms. Wajiha Mughal
Supervisor

Signature: _____

Dated: _____

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Group Members

Signature

1. Muhammad Abdul Rafay

2. Muhammad Mudassar Hassan

3. Muhammad Altahash Butt

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Chapter 1
INTRODUCTION

1.1 Introduction

The "Automated Transcript System" at the National University of Modern Languages [\[1\]](#) is a cutting-edge solution set to revolutionize how NUML [\[2\]](#) manages student records and transcripts. Tailored for students, coordinators, and the exam branch, this system promises enhanced efficiency and precision in academic record management.

For NUML coordinators, the system simplifies the management of student data, including subjects and credit hours. Automated GPA calculations and pass/fail determinations streamline administrative tasks, while real-time updates for subject-specific marks ensure precise academic progress tracking.

The NUML exam branch can seamlessly generate complete and incomplete transcripts, providing a comprehensive view of student performance throughout their academic journey.

Empowering NUML students with secure access to their academic records, the "Automated Transcript System" enables them to monitor progress and request transcripts, fostering transparency and self-service.

This innovative system marks a significant leap forward for NUML, promising increased efficiency, data accuracy, and convenience, delivering improved record management and an enhanced academic experience.

1.2 Motivation for Developed System

The motivation behind this project is to improve operational efficiency and reduce resource overhead within the exams branch. Currently, manual transcript management requires dedicated personnel for each department, consuming time and human resources. With the new system, we aim to automate transcript generation, saving time, reducing errors, and optimizing staff utilization. By modernizing record management, NUML will achieve streamlined, transparent, and accurate academic records, benefiting both the institution and its students.

1.3 Problem Statement

The National University of Modern Languages (NUML) confronts a multifaceted challenge in its current academic record management processes. NUML's manual transcript system relies on dedicated staff for each department, placing a substantial burden on the exams branch. This laborious method is riddled with errors, time delays, and an overwhelming reliance on human

resources. Consequently, students experience difficulties in accessing their semester results and transcripts promptly.

Previously, students had to physically visit the exams branch or rely on PDF documents sent by coordinators, which contained records for entire departments. The result: cumbersome, time-consuming processes and a lack of immediate access to individual academic performance data.

The solution lies in the "Automated Transcript System," which transforms this scenario. With this system, students can swiftly generate unofficial transcripts and view their results, saving them valuable time and hassle. Coordinators gain a user-friendly tool for streamlined record management. For the exams branch, the system facilitates one-click generation of official transcripts, reducing manual effort significantly.

1.4 Developed System Objective

To create a web application that can update and optimize bachelor student's academic record management at NUML. This system is intended to automate transcript creation, improve data accuracy, and minimize resource reliance within the examinations branch. Additionally, it gives students easy access to their academic records. Our project's aims include streamlining administrative operations, improving the overall student experience, and increasing the efficiency of NUML's academic record management system.

1.5 Developed System

The Automated Transcript System web service allows bachelor's students to conveniently access their academic information, create unofficial transcripts, and follow their progress. Coordinators can effectively handle student data, subjects, and grades, while the test branch can issue official transcripts. Administrators can manage user accounts and system settings. This simplified platform improves academic record administration at NUML by increasing transparency, reducing human work, and promoting correctness.

1.5.1 Features

Following will be the main features for our developed system.

- Web Application
- User authentication and access control.
- Student profile management.
- Subject and credit hour management.
- Pass/fail determination.

- Instant official and unofficial transcript generation.
- Academic progress tracking.
- Official transcript request functionality.

1.6 Scope for Developed System

Developed system fulfills the following scope:

- A web application for university management and students.
- The scope of the system includes the automation of transcript generation.
- It encompasses the management of student records by coordinators.
- The system handles academic data entry and calculations.
- It provides students with access to their academic progress.
- It allows for the generation of both official and unofficial transcripts.
- It streamlines the transcript generation operations in the exams branch.

1.7 Tools & Technologies for Developed System

There are the following required tools/technologies.

1.7.1 Software

- Visual Studio (.NET)
- Adobe Illustrator (For creating Logos, Pictures, etc.)
- Figma (For User Interface Design)

1.7.2 Hardware

- *Table 1.1* shows the tools we will need to create our developed system.

Table 1.1: Tools and Technologies used for developing IDC

Tools & Technologies	Version	Rationale
Microsoft Visual Studio	Microsoft Visual Studio Community 2022 (64-bit) V 17.5.4	To develop application
Microsoft .NET / ASP.NET MVC Core and Web Tools	V 17.5.318.41597	For making web application
Microsoft SQL Server Management Studio	V 17.9.1	For managing database

1.8 Milestones

By using Iterative Waterfall Model, we have plans shown in *Figure 1.1* to go on for the development of our developed system.

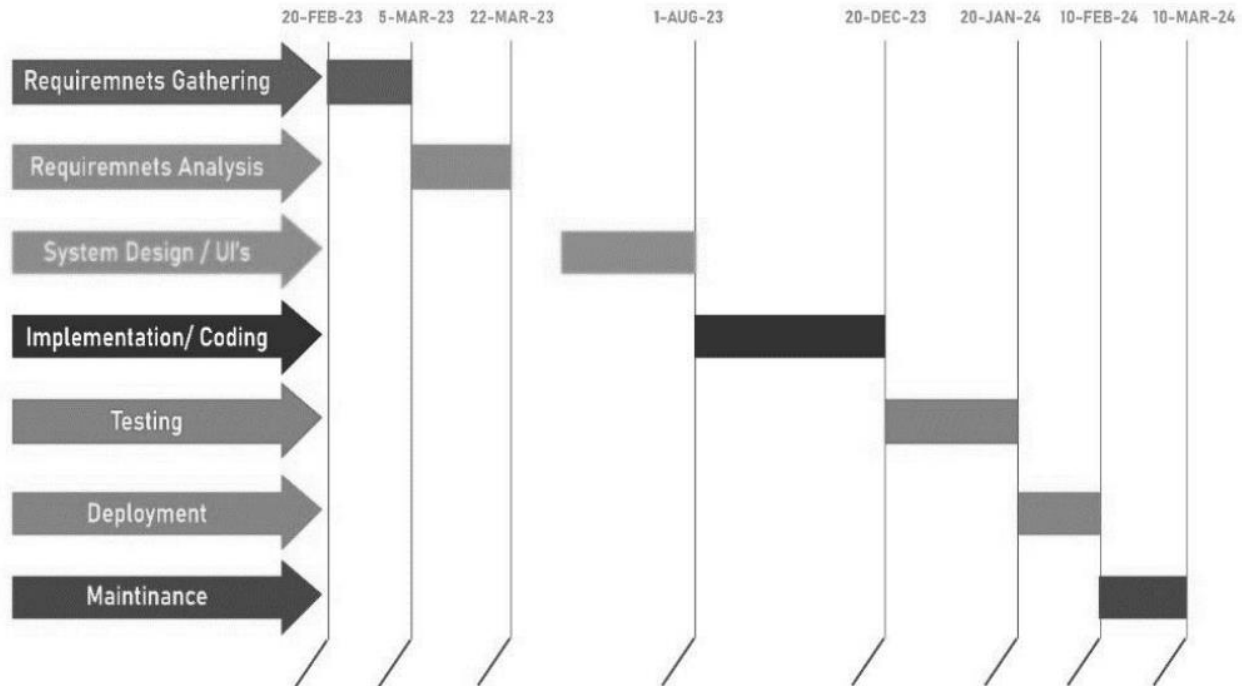


Figure 1.1: Gantt Chart for Automated Transcript System development lifecycle

1.9 Summary

In this chapter, we have given a little introduction of developed system. And also discussed some points like objective, motivation behind this project, scope of the project. We also mentioned milestones and some features of the project.

Chapter 2

BACKGROUND AND EXISTING SYSTEM

2.1 Introduction

This Chapter includes the features and the modules of proposed system, brief introduction to the existing systems already available in the market for analyzing human emotion, and also this chapter enables us to compare between the existing systems and our proposed system via their features.

2.2 Existing Systems

NUML's existing transcript system has been in place for many years, providing students with academic records and marks. Despite being a manual method, the existing transcript system has successfully produced both incomplete and complete DMCs for students. The manual approach also allowed for some customization, since clerks and coordinators could manually enter and check student data. However, as the institution expanded and the number of students rose, the manual approach became more time-consuming and prone to mistakes.

2.2.1 Transcript Maker

Transcript Maker [\[3\]](#) is a user-friendly online platform that simplifies the creation and management of high school transcripts. It provides a comprehensive toolset for students, parents, and educators to efficiently generate accurate transcripts, ensuring a hassle-free experience for academic record-keeping and college applications.

2.2.1.1 Features

- Custom transcript creation.
- User-friendly interface.
- Multiple transcript templates.
- Automated GPA calculation.
- Secure data storage.
- Digital transcript delivery.

2.2.1.2 Limitation

- Limited customization options.
- Cost considerations for certain features or templates.
- The accuracy depends on the user data input.
- Dependency on a stable internet connection.
- Variability in updates and customer support.

2.2.2 Parchment

Parchment [\[4\]](#) is an online platform that simplifies the process of requesting and sending academic transcripts securely. Students and alumni can conveniently order both digital and physical transcripts through this service. Parchment often offers a tracking system that enables users to monitor the status and delivery of their transcripts, ensuring a streamlined and transparent experience. The platform commonly integrates with educational institutions, facilitating the electronic delivery of transcripts to receiving institutions. Additionally, some organizations rely on Parchment for third-party verification of academic credentials, assuring the authenticity of transcripts. With a user-friendly interface, Parchment aims to enhance the efficiency of transcript management for educational institutions and individuals alike.

2.2.2.1 Features

- Secure online transcript ordering.
- Digital and physical transcript options.
- Transcript tracking for users.
- Integration with educational institutions.
- Third-party verification of academic credentials.
- User-friendly interface.

2.2.2.2 Limitation

- Possible fees for transcript requests.
- Availability depends on educational institution partnerships.
- Data privacy considerations.
- Variability in user support quality.
- Feature and update availability may vary by institution.

2.2.3 Ellucian Banner

Ellucian Banner [\[5\]](#) is a sophisticated Enterprise Resource Planning (ERP) solution tailored to higher education organizations. It acts as a single center for coordinating multiple administrative duties inside colleges and universities, allowing institutions to simplify operations and enhance efficiency.

2.2.3.1 Features

- Student Information System (SIS) for managing student records.
- Finance and accounting tools for budgeting and financial reporting.

- Human resources management, including payroll and benefits administration.
- Extensive reporting and analytics capabilities.
- Student engagement features like self-service portals.

2.2.3.2 Limitation

- Complexity in implementation and maintenance.
- Cost may be a significant factor for smaller institutions.
- Staff may require training to maximize the system's potential.

2.2.4 Comparison Between Existing System & Developed System

In *Table 2.1* we can see the comparison of features between existing system and developed system shown in detail.

Table 2.1: Comparison between existing systems and developed system

Features	Transcript Maker Tool	Parchment Tool	Ellucian Banner Tool	Proposed system
Automatic processing of transcript	No	Yes	No	Yes
Student result management	No	No	Yes	Yes
Customized option for transcript	Yes	Yes	No	Yes
Automatic calculation of GPA	Yes	No	Yes	Yes
Secure transcript and storage of accurate data	Yes	Yes	Yes	Yes

2.3 Modules of developed system

The modules of developed system are written below.

2.3.1 Coordinator Module

This module empowers coordinators to efficiently manage student records and handle the critical task of entering student marks. Coordinators can assign courses to specific semesters, add or update student marks, and ensure the accuracy of academic records within the system. The system integrates automated GPA calculation based on the marks entered, streamlining the process of academic evaluation. Additionally, the module includes logic to analyze student performance, determining whether a student meets the criteria for promotion to the next academic level or if they have failed.

2.3.2 Exam Branch Module

This module enables authorized staff to search for and generate official transcripts using student system IDs or roll numbers. The system generates complete DMCs for graduated students and incomplete DMCs for those still studying. Also, they can check out the results for the students and issue their result for specific semesters.

2.3.3 Student Module

This module allows students to access their academic records and track their progress throughout their academic career. The system generates an unofficial transcript that shows the courses taken, grades received, and cumulative GPA. Students can also view their previous semester result. This accessibility ensures that students can stay informed about their performance and make informed decisions regarding their academic future. Additionally, the ability to download their unofficial transcript allows for easy sharing of their academic achievements.

2.3.4 Admin Module

The admin module in the system plays a pivotal role in system management and user administration. Administrators can create and manage student profiles, generate unique student IDs, add and oversee coordinators, manage overall degree courses, , assigning subjects to specific semester according to their scheme of study. They have the authority to control user access, configure system settings, and provide user support. Additionally, administrators contribute to system security and compliance, ensuring the smooth functioning of the automated transcript system.

2.4 Flow Diagram of Developed System

In Figure 2.1 we can see the flow of the developed system which shows us all the modules.

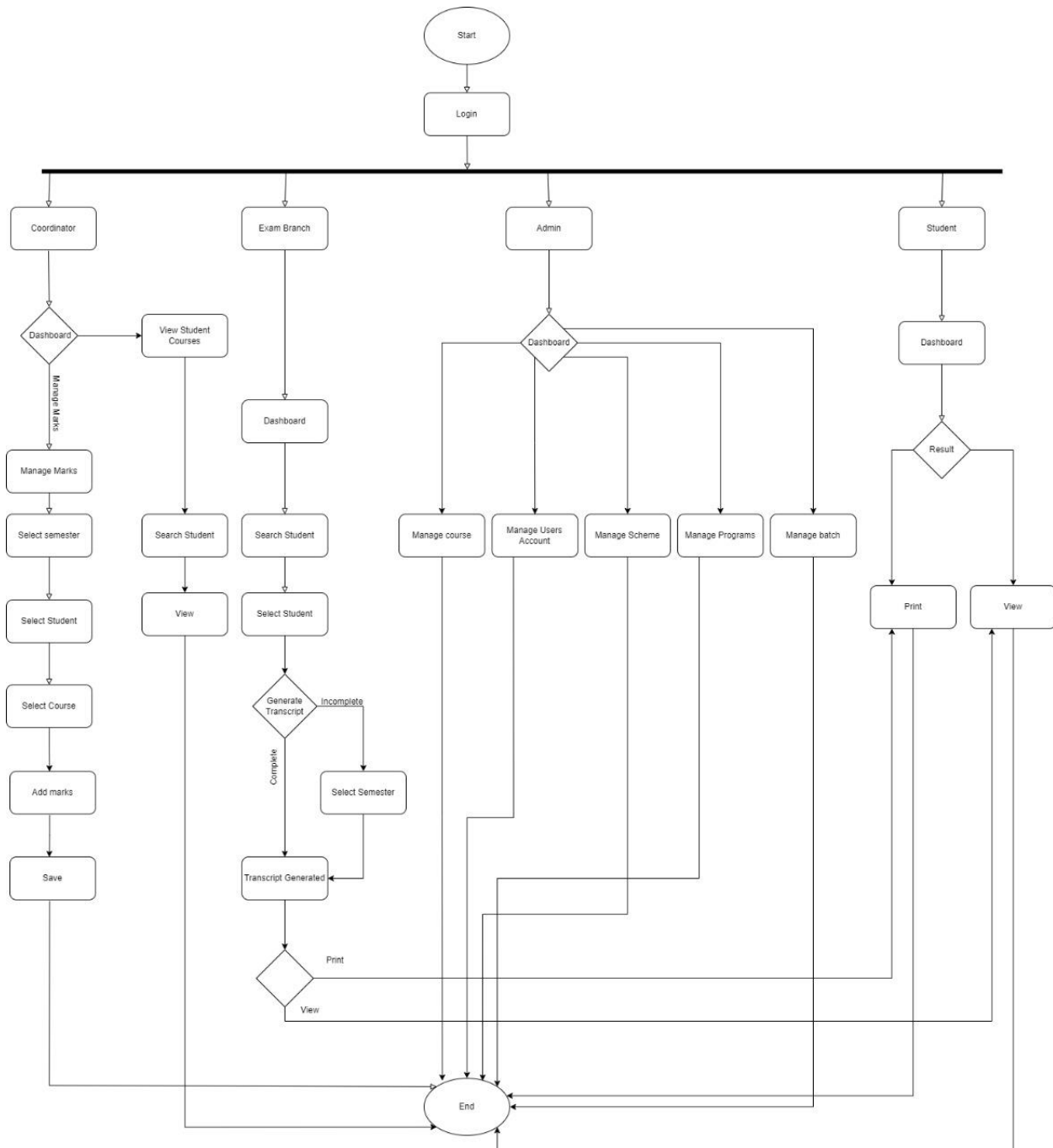


Figure 2.1: Flow Diagram of Automated Transcript System

2.5 Methodology Used for Developed System

The Iterative Waterfall Model is a project management approach that combines the structured, sequential phases of the traditional Waterfall model with iterative cycles for feedback and adaptation. It offers a middle ground between predictability and flexibility, allowing for course

corrections based on stakeholder feedback and evolving project requirements. This approach ensures well-defined milestones and comprehensive documentation while also addressing potential risks early in the project lifecycle. It's suitable for projects that require a balance between structured planning and adaptability.

In *Figure 2.2* we can see the Iterative WaterFall Approach which we have used for developing our system

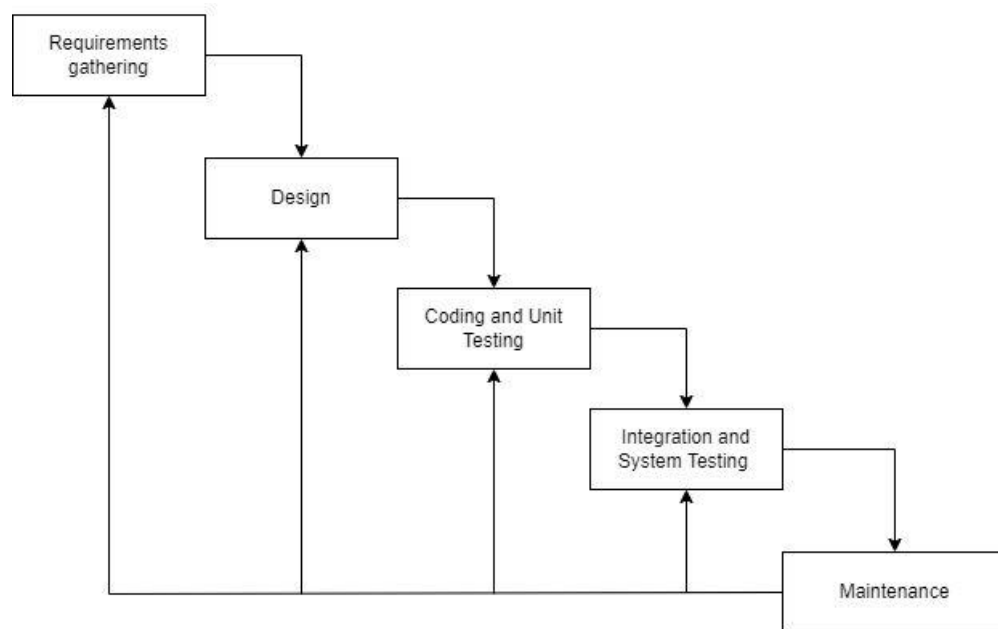


Figure 2.2: Model used to develop Automated Transcript System

2.6 Feasibility Study for Developed System

In advance of implementation, the designed system's feasibility is determined from multiple perspectives in a feasibility study. The information that follows provides an outline of the feasibility study main components.

2.6.1 Technical Feasibility

Evaluate the technical capabilities and resources required to develop and maintain the system. Assess the availability of hardware, software, and infrastructure to support the system. Consider any technical challenges, risks, or dependencies that may impact system development and performance. By providing the option to download their unofficial transcript, the system facilitates easy sharing of their academic accomplishments with potential employers, graduate programs, or other institutions. This feature supports students in managing and showcasing their academic progress effectively, contributing to their overall success and future opportunities.

2.6.2 Operational Feasibility

Assess how well the system aligns with the organization's operational goals and objectives. Evaluate the impact of the system on existing business processes and workflows. Identify potential changes or adjustments required in operational practices to accommodate the new system.

2.6.3 Resource Feasibility

Assess the availability of human resources, particularly the skilled professionals needed for system development, implementation, and support. Consider any training requirements for employees to efficiently utilize and manage the technology.

2.6.4 Risk Assessment:

Identify potential risks and uncertainties that could impact the success of the system. Develop risk mitigation strategies to address and minimize these risks.

In *Table 2.2* we can see the resources which are used while developing the system.

Table 2.2: Resources and Requirements

Resource	Requirement	Rational
Internet	3G or above	For basic guidelines and help
Laptop/PC	8GB RAM or above	Core-i5-5 th Generation 2.20 GHz or faster processor 10 GB of available hard disk space, for development and maintenance of system
PC/ Laptop /Mobile	Any Laptop/ PC/Mobile	For using web application

2.7 Summary

In this chapter, we have given details of previous systems which is related to developed system. We mentioned some of their pros and cons. And given a detailed comparison between all existing systems and developed system in this chapter. We also introduced the methodology that will use in our project.

Chapter 3

REQUIREMENT SPECIFICATION

3.1 Introduction

This chapter is all about the requirements of the system. It deals with requirements analysis as well as documentation of these requirements. It covers the ‘What’ part of the system. In this chapter, system modelling is done and use case diagram is designed. System modelling is the prediction of different views of the system by developing abstract models of the system. Every model, so far, represents a specific angle, view or perspective of that system. System modelling of the system is done here by using graphical notations which are based on notations in the Unified Modelling Language (UML). System modelling provides an analyst comprehensive understanding of the system, and it helps the developer in understanding the flow of working of software and modelling approaches, which are used to interact with users. In this regard, system design explains the environment in which the system works. The method of modeling is helpful for the progress of the system as it gives a clear picture of the system’s working, and it also caters the vulnerability of the user in the beginning phase of system’s lifecycle.

3.2 Functional Requirements

Functional requirements are product features or functions that developers must implement to enable users to accomplish their tasks. So, it’s important to make them clear both for the development team and the stakeholders. Generally, functional requirements describe system behavior under specific conditions. For example, we can group them on the basis of the functions a given feature must perform in the end product.

The functional requirements of the system are mentioned below.

3.2.1 Login

In *Table 3.1* the functional requirements for Login functionality are defined.

Table 3.1: Functional Requirement for Login

Identifier	FR-01
Title	Login
Requirements	The system shall provide secure login using username and password authentication.
Priority	High

3.2.2 GPA Calculation

In *Table 3.2* the functional requirements for GPA Calculation functionality are defined.

Table 3.2: Functional Requirement for GPA Calculation

Identifier	FR-02
Title	GPA Calculation
Requirements	The system shall automatically calculate the GPA of students and, based on semester report.
Priority	High

3.2.3 Access to Academic Records

In *Table 3.3* the functional requirements for Academic Records Access are defined.

Table 3.3: Functional Requirement for Access to Academic Records

Identifier	FR-03
Title	Access to Academic Records
Requirements	Registered students shall have access to their academic records, including semester results and GPAs.
Priority	High

3.2.4 Generate Unofficial Transcripts

In *Table 3.4* the functional requirements for Unofficial Transcripts are defined.

Table 3.4: Functional Requirement for Unofficial Transcripts

Identifier	FR-04
Title	Generate Unofficial Transcripts
Requirements	Students shall have the ability to generate unofficial transcripts, including current and past semester performance.
Priority	High

3.2.5 Generate Official Transcripts

In *Table 3.5* the functional requirements for Official Transcript Generation are defined.

Table 3.5: Functional Requirement for Generate Official Transcripts

Identifier	FR-05
Title	Generate Official Transcripts
Requirements	The system shall enable the generation of official transcripts for graduate students, including complete and incomplete transcripts as required.
Priority	High

3.2.6 User Account Management

In *Table 3.6* the functional requirements for User Account Management are defined.

Table 3.6: Functional Requirement for User Account Management

Identifier	FR-06
Title	User Account Management
Requirements	Administrators shall allow the management of user accounts.
Priority	High

3.2.7 Degree Course Management

In *Table 3.7* the functional requirements for Course Management are defined.

Table 3.7: Functional Requirement for Course Management

Identifier	FR-07
Title	Degree Course Management
Requirements	Administrators shall facilitate the management of overall degree courses, including addition, update, and removal.
Priority	High

3.2.8 Course Details Definition

In *Table 3.8* the functional requirements for Course Details are defined.

Table 3.8: Functional Requirement for Course Details Definition

Identifier	FR-08
Title	Course Details Definition
Requirements	Administrators shall be able to define course details such as course names and credit hours
Priority	High

3.2.9 Marks Input and Update

In *Table 3.9* the functional requirements for Marks insertion and updating are defined.

Table 3.9: Functional Requirement for Marks input and update

Identifier	FR-09
Title	Marks Input and Update
Requirements	Coordinators shall input and update subject-specific marks.
Priority	High

3.2.10 Assign courses to semesters

In *Table 3.10* the functional requirements for assigning courses to semester are defined.

Table 3.10: Functional requirement for Assigning courses to semesters

Identifier	FR-10
Title	Assign courses to semesters
Requirements	Coordinators shall have the capability to assign courses to specific semesters within the academic program.
Priority	High

3.3 Non-Functional Requirements

Non-functional requirements are the specifications that illustrate the system's operational abilities and limitations that augment its functionality. These requirements include usability and portability. Some of the non-functional requirement covered by our project are stated below:

3.3.1 Performance

- Ensure that the system provides fast access to academic records, responding to user queries within 3 seconds.

3.3.2 Security

- Implement strong security measures, including encryption for data transmission and secure password handling.
- Restrict access to data based on user roles, ensuring that users can only access relevant information.

3.3.3 Scalability

- Design the system to easily expand as user numbers grow or new departments are added.
- Ensure that the database can store at least 10 years of academic data.

3.3.4 Availability

- Guarantee a high system uptime of at least 99.9% during regular operating hours.

3.3.5 Reliability

- Implement automated backup mechanisms to protect against data loss or system failures.
- Maintain data consistency and accuracy, with a low allowable data error rate.

3.3.6 Usability

- Create a user-friendly and accessible interface that follows best practices.

3.4 Use Case

Use case diagram are an essential tool for requirements analysis and software design, as they provide a clear and concise representation of a system's functionality and how it interacts with users and other systems. Use case diagrams can help identify potential issues or gaps in the system's functionality, which can be addressed before the system is built or released. In our system, there are only four actors i.e., Student, Coordinator, Admin, Exam Branch Staff. The *Figure 3.1* shows the use case diagram of our system

3.4.1 Use Case Diagram for Automated Transcript System

In Figure 3.1 the Use Case diagram for the developed system is Illustrated.

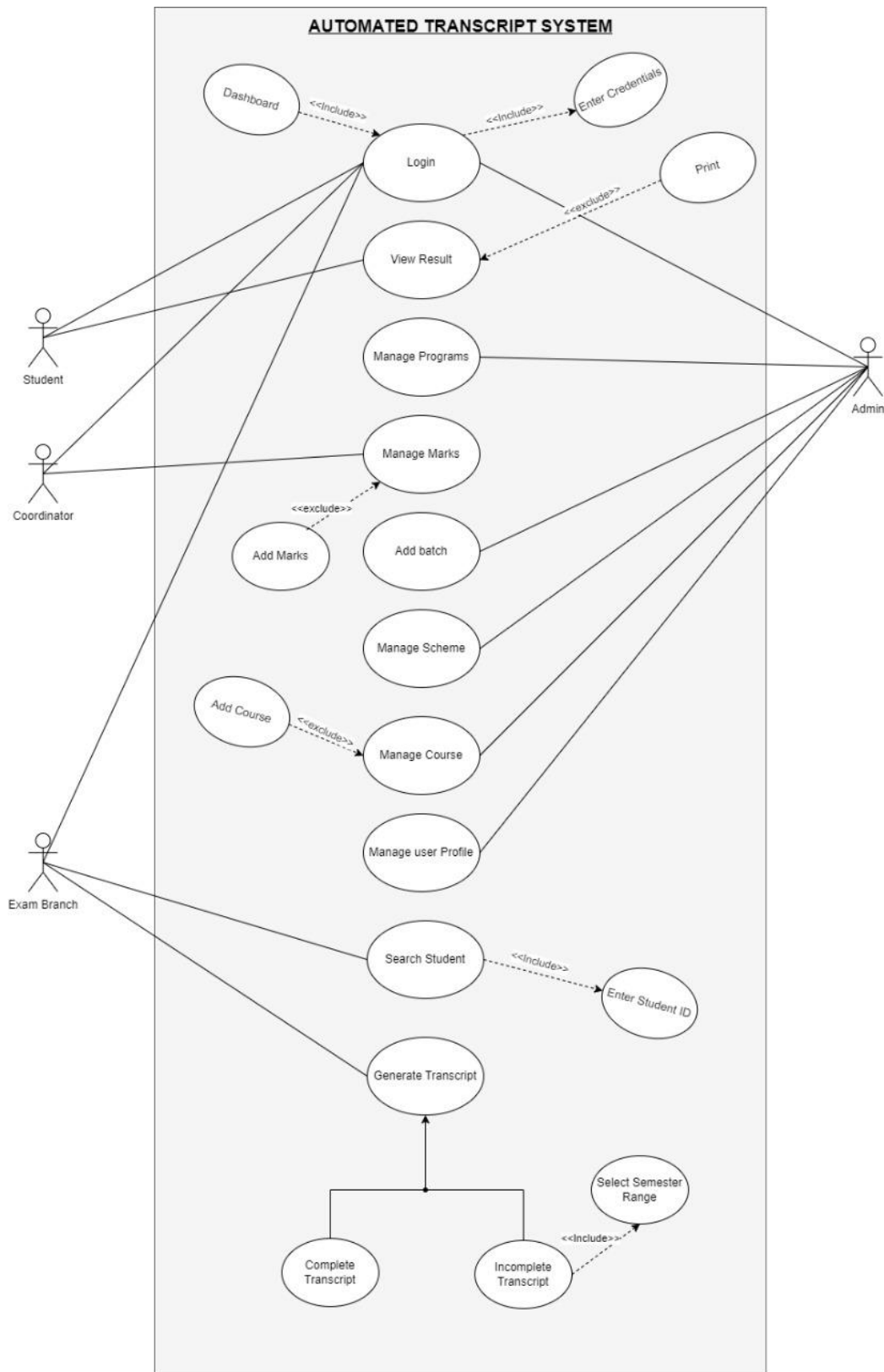


Figure 3.1: Use case diagram for Automated Transcript System

3.4.2 Functional Requirements as Use Cases

The interaction between external actors of the system with the system itself is known as a use case. The use case describes the function and actions to be performed by the actors of the system. Use cases are the compendium of sequences, which consist of every feasible state of software for the flow of performance control as a result comprising of all attainable performances, which a system could do. These interactions help in understanding the system's behavior in various scenarios and are essential for defining the system's functional requirements in a structured manner. By detailing how users and other systems will interact with the system, use cases ensure that all necessary functions are captured and accurately described. This approach also aids in identifying potential gaps or issues in system functionality early in the development process. Additionally, use cases provide a clear, user-oriented perspective on system requirements, facilitating better communication and alignment between stakeholders and the development team. The use case descriptions for our Automated Transcript System outline each user interaction and the corresponding system response, ensuring comprehensive coverage of all functional requirements and guiding the development and testing phases effectively. Properly documented use cases also serve as a reference point for validating the system's functionality and ensuring it meets the specified requirements.

3.4.3 Use Case Description

Use case description explains the use case diagram by detailing how actors interact with the system. It provides a narrative that outlines the steps involved in each use case, including the inputs, interactions, and outputs. This description helps in understanding the functionality from the user's perspective, ensuring that all scenarios are covered comprehensively. Use case descriptions are crucial for validating that the system meets the needs of its users and for identifying any potential improvements or modifications. They serve as a reference for developers and testers, guiding them in implementing and verifying the system's functionality. The use case descriptions for the system are provided below, illustrating the interactions between different actors and the system's response to their actions. The descriptions ensure that all possible use scenarios are covered and provide a clear roadmap for system implementation and testing. The use case descriptions for our Automated Transcript System are outlined below, capturing the essence of each user interaction and the system's responses to ensure comprehensive functionality. Moreover, they help in refining the system design by offering insights into real-world usage and potential areas for improvement.

3.4.3.1 Use Case Description for Login

Table 3.11 shows the use case description for login

Table 3.11: Use case description for Login

Use Case ID	UC 01
Use Case Name	Login
Actor	Student/Coordinator/Admin/Exam Branch
Trigger	-
Description	Login into the system
Pre-Conditions	Account Registered
Main Success Scenario	
User Action	System Response
1) User Open Web portal	1.1) System Display Login Page
2) User Enter the Credentials	
3) User Press Login	3.1) System Verifies the Credentials and Check Type of User
4) User is Redirected to Homepage according to its Type.	
Post Condition: The user is Logged into the portal	
Alternative Flow: 3.1 System Detects incorrect Credentials	

3.4.3.2 Use Case Description for Generate Profile

Table 3.12 shows the use case description for generate profile.

Table 3.12: Use case description for Generate Profile

Use Case ID	UC 02
Use Case Name	Generate Profile
Actor	Admin
Trigger	New Users
Description	Admin will create accounts for All users including student, coordinator and exam branch
Pre-Conditions	The Admin Should Login Into System
Main Success Scenario	
User Action	System Response
1) Admin Enter User Details	
2) Admin click on create account	2.1) System creates a new user account
Post Condition: The user account is created	
Alternative Flow: 2. Display Error Message	

3.4.3.3 Use Case Description for View Result

Table 3.13 shows the use case description for view result.

Table 3.13: Use case description for View Result

Use Case ID	UC 03
Use Case Name	View Result
Actor	Student
Trigger	-
Description	The Interacting Actors can View Results.
Pre-Conditions	Account logged in
Main Success Scenario	
User Action	System Response
1) Go to results	
2) Press View	2.1) Results of semester will appear
Post Condition: The Result will be Displayed to the Actor.	
Alternative Flow: 1. Logout.	

3.4.3.4 Use Case Description for generate Unofficial Transcript

Table 3.14 shows the use case description for generate unofficial transcript

Table 3.14: Use case description for Generate Unofficial Transcript

Use Case ID	UC 04
Use Case Name	Generate Transcript
Actor	Exam Branch
Trigger	Student want his official transcript
Description	Exam branch can generate transcript
Pre-Conditions	Exam branch person should be logged in system
Main Success Scenario	
User Action	System Response
1) Search student ID	
2) Press generate	2.1) System will generate transcripts
Post Condition: The Transcript will be Displayed or Printed to the Actor.	
Alternative Flow: 1.1 Student not found 2.1 Incomplete degree status	

3.4.3.5 Use Case Description for Manage Courses

Table 3.15 shows the use case description for Manage Course.

Table 3.15: Use case description for Manage Course

Use Case ID	UC 05
Use Case Name	Manage Course
Actor	Admin
Trigger	-
Description	The courses offered in the semester are added/managed
Pre-Conditions	Valid Course is Created and has required Details related to them.
Main Success Scenario	
User Action	System Response
1) View Course Details	1.1) Display Courses
2) Edit/Update Courses Details	2.1) Update/Modify Course Details
Post Condition: The Course Details are Updated and Stored Successfully in Database.	
Alternative Flow: 2. Enter Invalid Course Details	

3.4.3.6 Use Case Description for Search Student Details

Table 3.16 shows the use case description to search student details.

Table 3.16: Use case description for Search Student Details

Use Case ID	UC 06
Use Case Name	Search Student Details
Actor	Exam Branch
Trigger	-
Description	Exam Branch can search for student academic record
Pre-Conditions	Account logged in
Main Success Scenario	
User Action	System Response
1) Search Student by ID	1.1) Display Student record
2) Validate and Verify Student Records and Generate Transcript	
Post Condition: The Exam Branch gets the Transcript	
Alternative Flow: 1. The Student ID is Invalid 2. The Record is Missing for any specific Student	

3.5 Summary

In this chapter, we introduced requirements of developed system like some functional and non-functional requirements. Also, we create some use cases which will show the working of modules that exists in developed system.

Chapter 4
SYSTEM MODELING

4.1 Introduction

In this chapter, the modeling of the developed system is performed. The process of developing abstract models of a system is known as system modeling, with every model shows a different view or perspective of the system. System modeling here is to indicate this system using some kind of graphical notation, which is based on notations in the Unified Modeling Language.

The viewers and analyst can understand the working of the system and models approaches which are used to communicate with the customers with the help of system modeling. The system design shows the characteristics and also the environment where the system will work. By visualizing the system through different types of diagrams, such as use case diagrams, activity diagrams, and sequence diagrams, system modeling facilitates a comprehensive understanding of the system's behavior and requirements. It also helps in identifying design flaws and ensuring that the system meets the intended objectives. This detailed modeling approach supports effective communication between developers, stakeholders, and end-users, leading to a more accurate and user-centered system design.

4.2 Interface Design

Interface design is one the most important part of engineering process because user's overall experience depends on how well your interface is designed. No matter how well a system does its functionality but if user is unable to do a specific task because of non-conformed interface design, it will leave a bad impression. For our system, interface is designed for mobile and in such a way that it is easy for its users to understand and interact with. Interface design is a critical component of the engineering process, as the user's overall experience hinges on how intuitively and effectively the interface facilitates interaction and task completion.

4.2.1 High Fidelity Prototype

High fidelity prototypes often known as Hi-fi is computer-based prototypes of the interface. It gives an idea of how user interface will be designed. It has more aesthetics and working presentation that's why most of the requirement engineering processes gather user preferences then designs prototype types to give a general idea of how product will look like or sometimes it also shows working too. Now a days a lot of tools are available which allow you to design these prototypes with minimum human effort. High fidelity prototypes, or Hi-fi, are advanced, computer-based representations of the user interface that provide a detailed and realistic preview of the final product's appearance and functionality, leveraging modern tools to streamline the design process and capture user preferences more effectively.

4.2.1.1 Login Page

In *Figure 4.1* Login Screen for the Automated Transcript System is shown in Hi-Fi Prototypes.

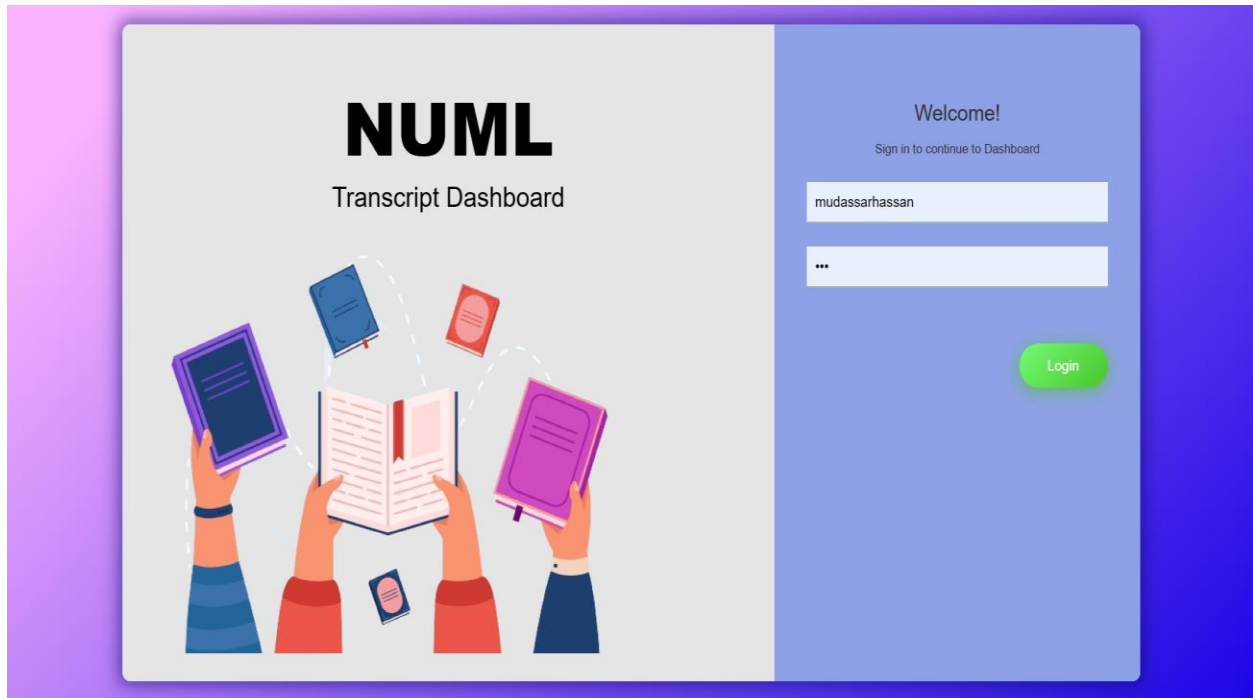


Figure 4.1: Login Screen

4.2.1.2 Exam Branch Dashboard

In *Figure 4.2* Exam Branch Dashboard for the Automated Transcript System is shown in Hi-Fi Prototypes.

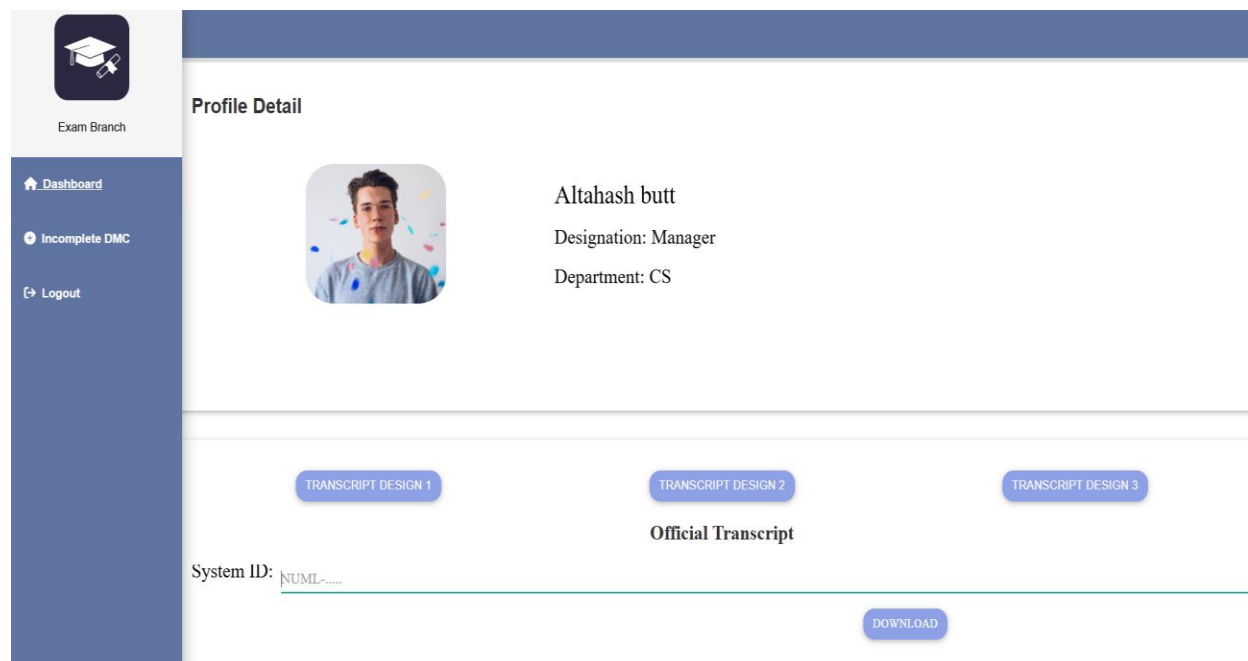


Figure 4.2: Exam Branch Dashboard

4.2.1.3 Student Dashboard

In *Figure 4.3* Student Dashboard for the Automated Transcript System is shown in Hi-Fi Prototypes.

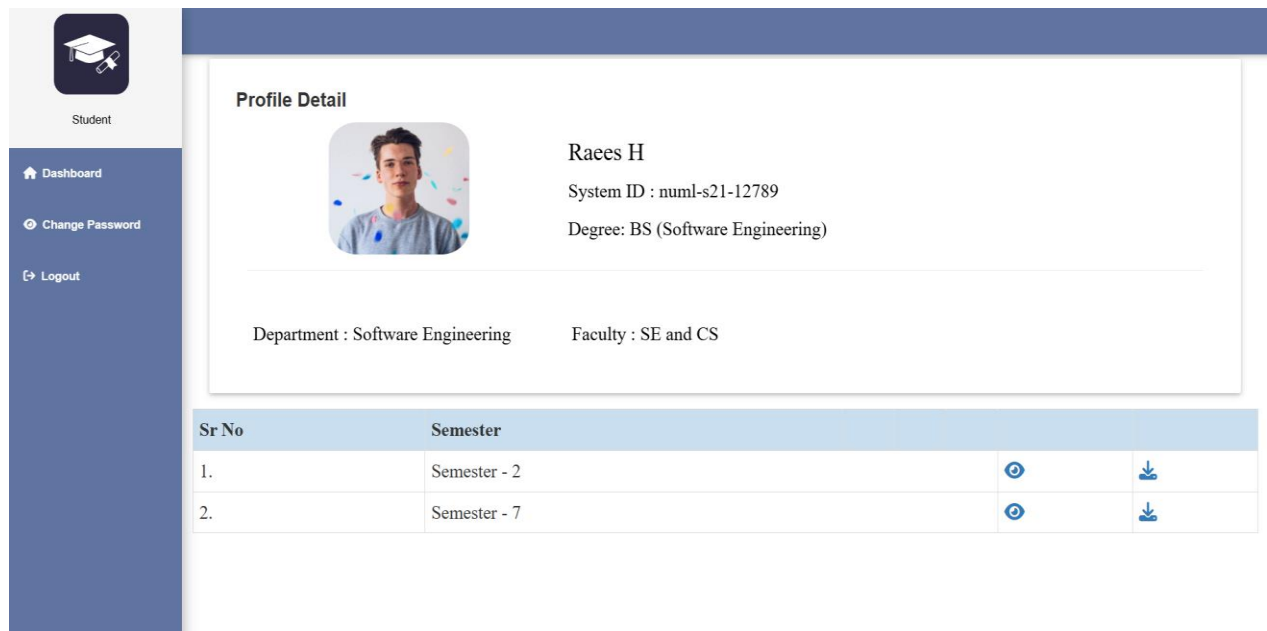


Figure 4.3: Student Dashboard

4.2.1.4 Coordinator Dashboard

In *Figure 4.1* Coordinator Dashboard for the Automated Transcript System is shown in Hi-Fi Prototypes.

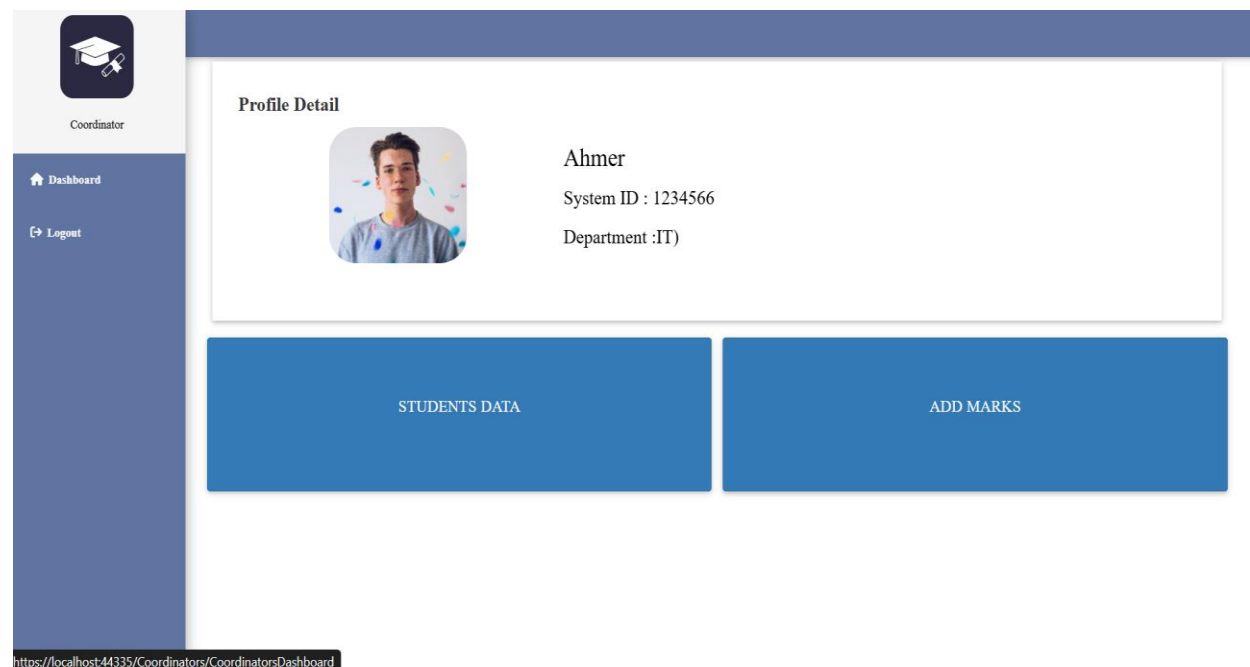


Figure 4.4: Coordinator Dashboard

4.2.1.5 Admin Dashboard

In *Figure 4.5* Admin Dashboard for the Automated Transcript System is shown in Hi-Fi Prototypes

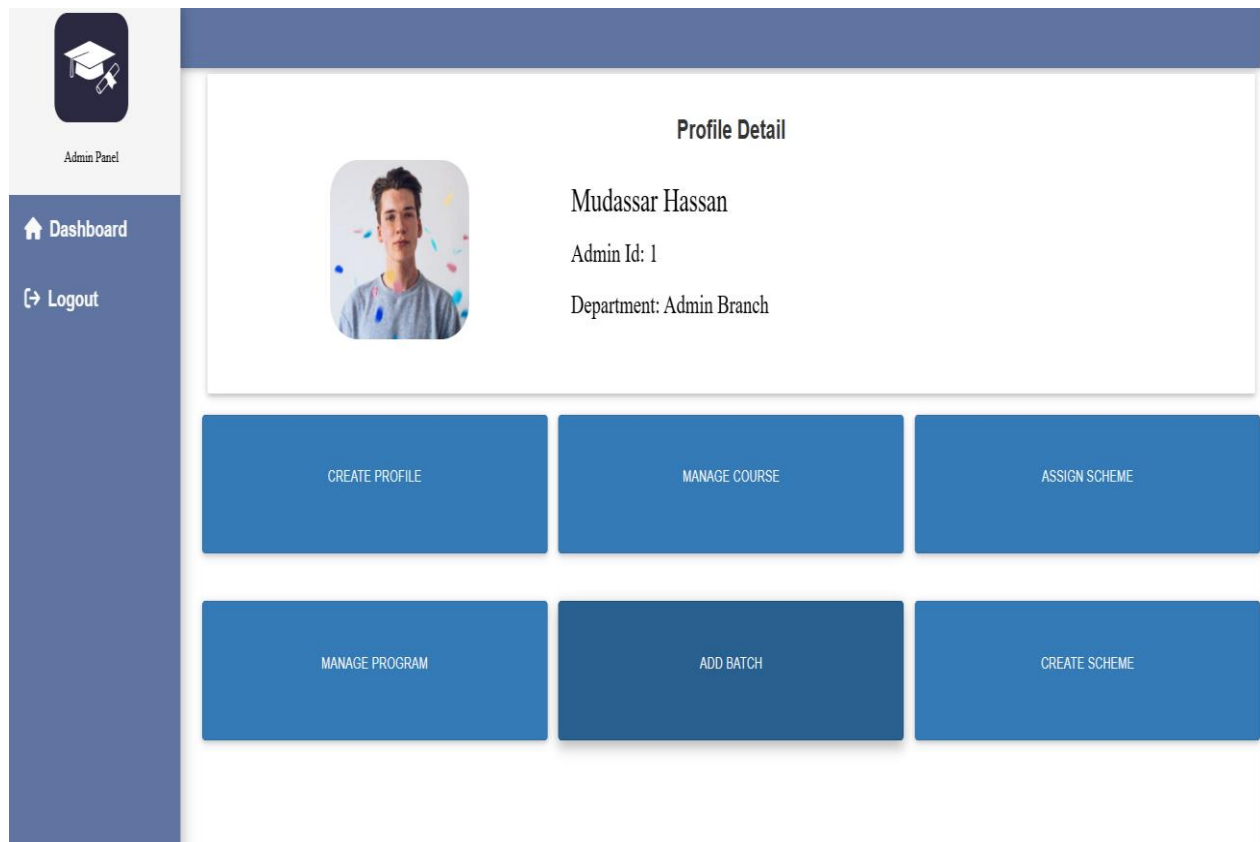


Figure 4.5: Admin Dashboard

4.3 (4+1) View Model

The (4+1) View Model is a robust framework that supports the comprehensive description of software architectures by incorporating multiple perspectives. The development view focuses on the system's modular structure, detailing the organization of software components and their interactions. The logical view represents the system's functional requirements, showcasing how the system fulfills its intended functionalities through various components. The physical view illustrates the system's deployment, emphasizing hardware configurations and network infrastructure. The process view addresses the dynamic aspects of the system, such as concurrency and data flow. By integrating these views with a use-case model, the (4+1) View Model provides a holistic understanding of the system, ensuring that all aspects of the architecture are thoroughly addressed and communicated to stakeholders. This comprehensive approach aids in identifying and resolving architectural issues early in the development cycle, leading to more reliable and maintainable systems.

In Figure 4.6 you can see the relation among different views.

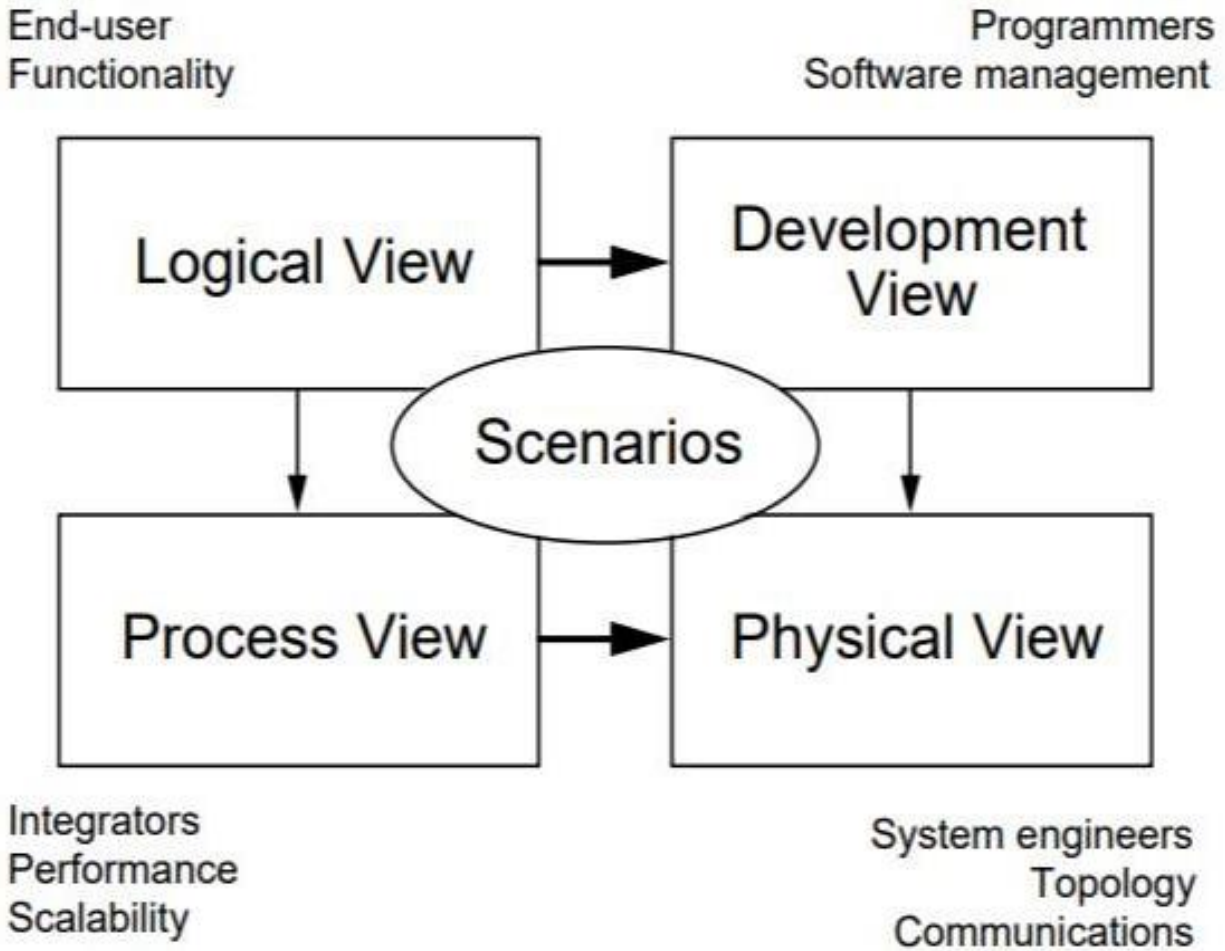


Figure 4.6: 4+1 View Model

4.4 Logical View

In logical view, we show static models view and its connection with functionalities of our projects. For this purpose, we use

1. Class diagram.

4.4.1 Class Diagram

The class diagram is extensively used by software developers during system development and design, as it outlines the system's classes, attributes, methods, and their relationships. This diagram not only provides a clear blueprint for the system's structure and functionality but also helps in understanding how different components interact with each other, thereby facilitating better planning and implementation of the system's architecture.

4.4.1.1 Class Diagram for Automated Transcript System

In Figure 4.7 you can see Class Diagram for Profile Generation of the System.

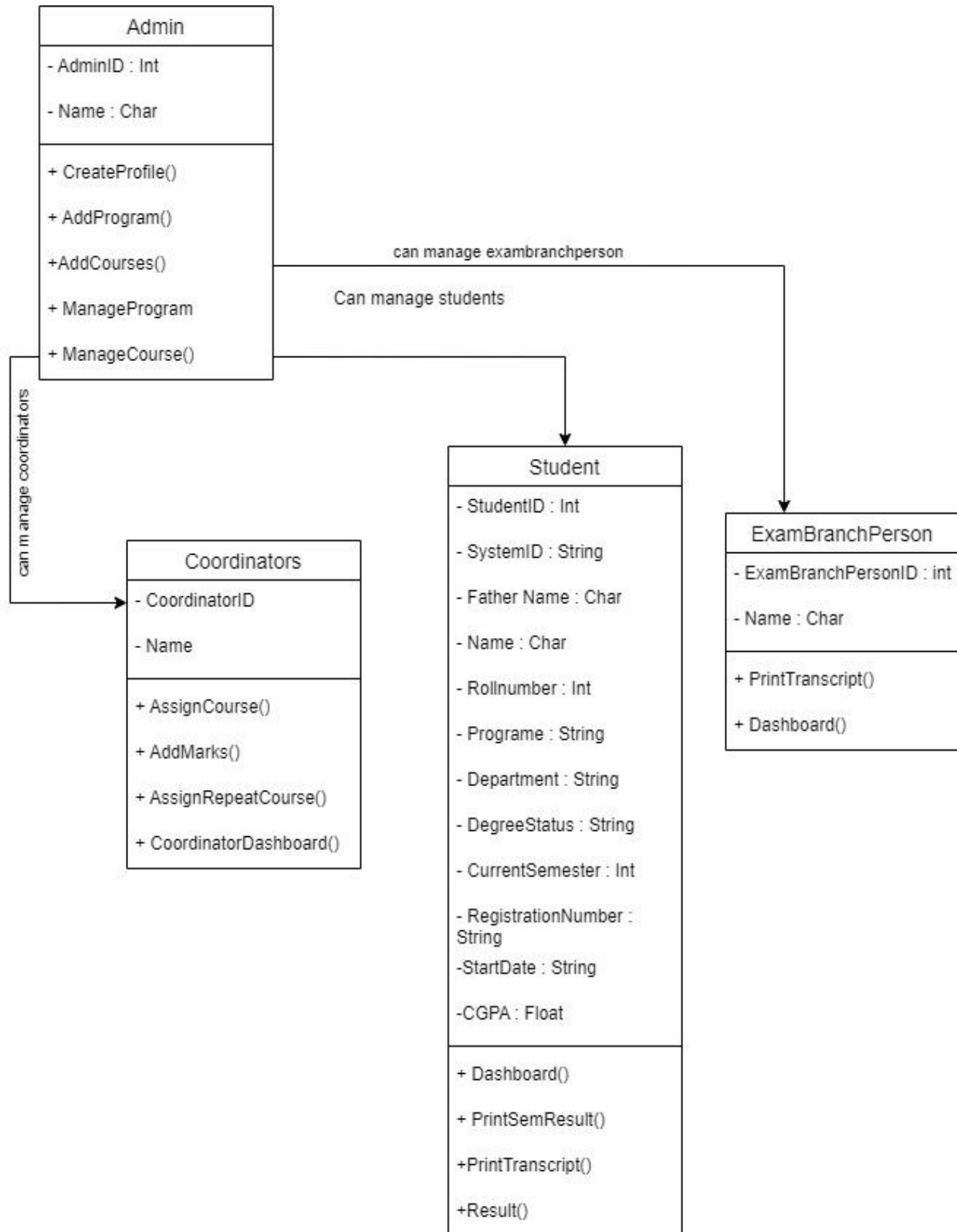


Figure 4.7: Class Diagram for Automated Transcript System

4.5 Process View

The process architecture used to deal with some non-functional requirements like performance and availability. It addresses concurrency, distribution, fault tolerance, system integrity, and how the main abstractions from the logical view fit into the process architecture.

The process view includes:

1. Activity diagram.
2. Sequence Diagram.

4.5.1 Activity Diagram

Activity diagram display the dynamic behavior of the system. An activity diagram is generally a flow chart that shows how one activity leads to another. It also deals with the dynamic behavior of the system by focusing on the runtime behavior of the system. Activity diagrams are similar as a flowchart except that it consists of activities performed by the system.

4.5.1.1 Activity diagram for Login

In Figure 4.8 you can see Activity Diagram for Login of the Automated Transcript System.

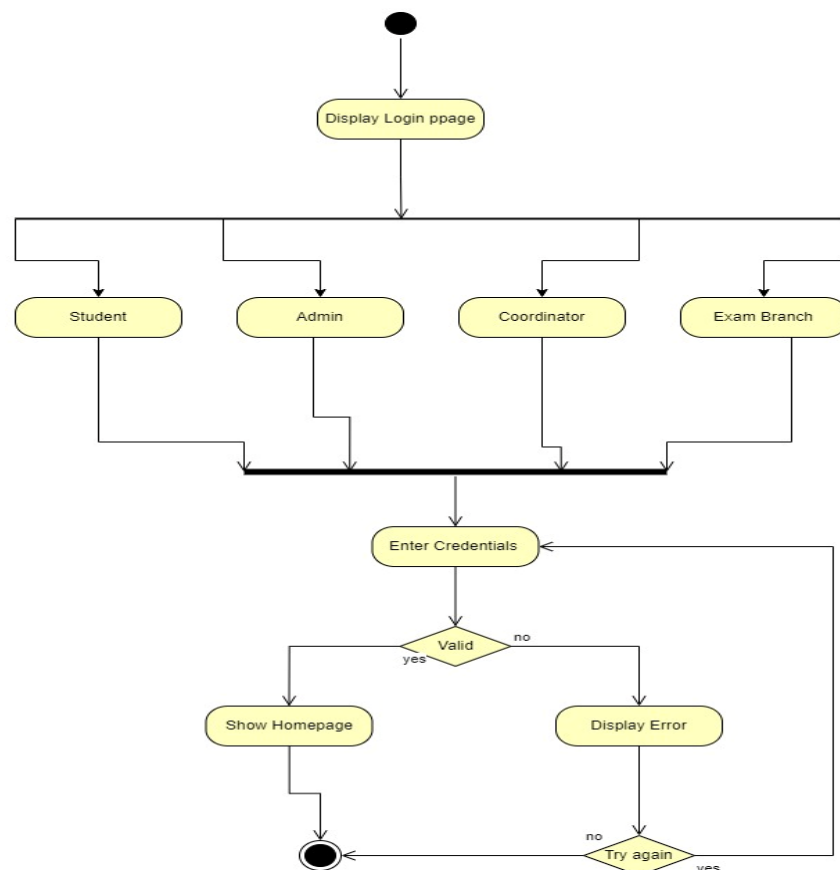


Figure 4.8: Activity Diagram for Login

4.5.1.2 Activity diagram for Manage Courses

In *Figure 4.9* you can see Activity Diagram for manage Courses of the Automated Transcript System.

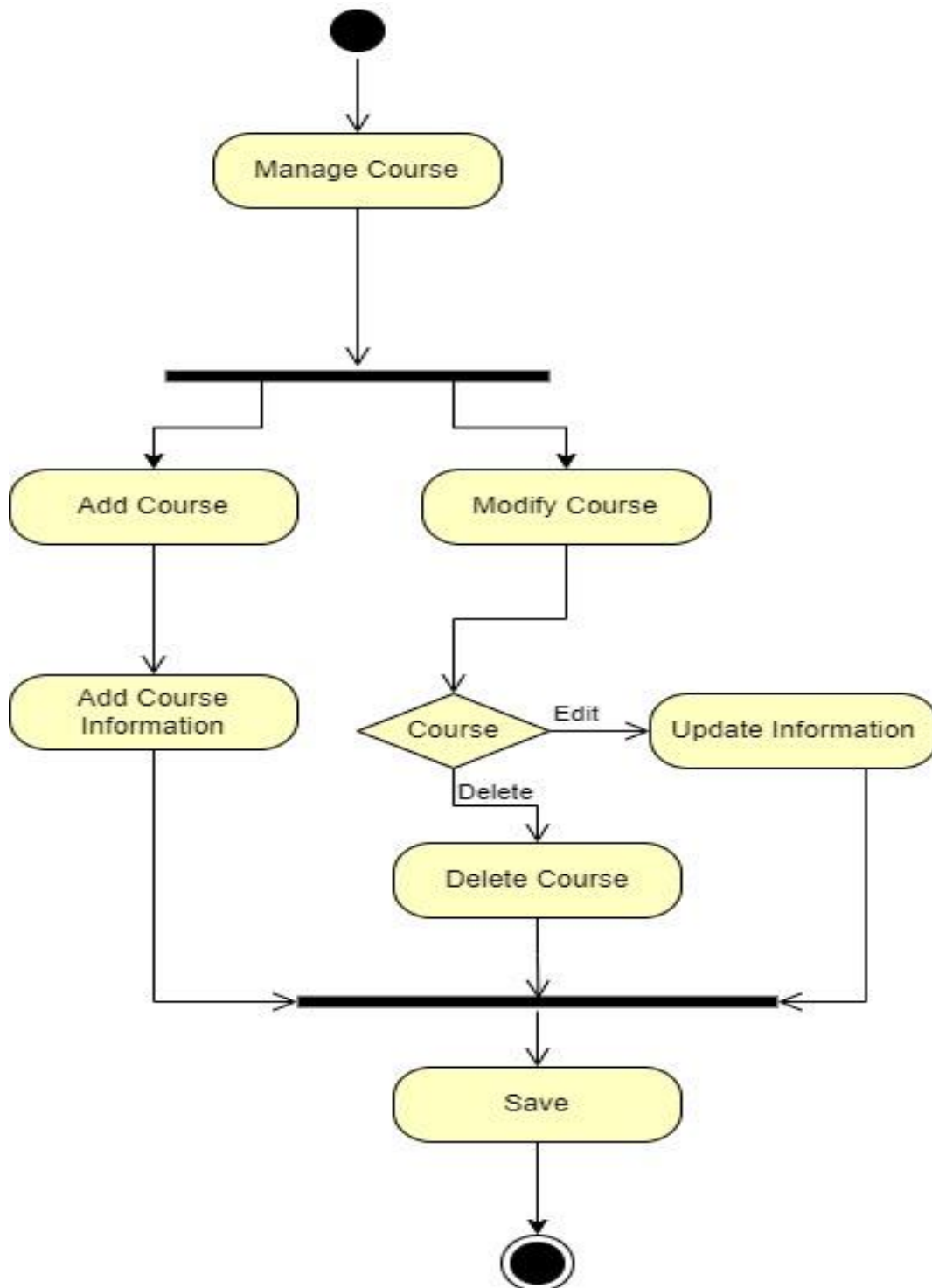


Figure 4.9: Activity Diagram for Manage Courses

4.5.1.3 Activity diagram for Transcript Generation

In *Figure 4.10* you can see Activity Diagram for Transcript Generation of the Automated Transcript System.

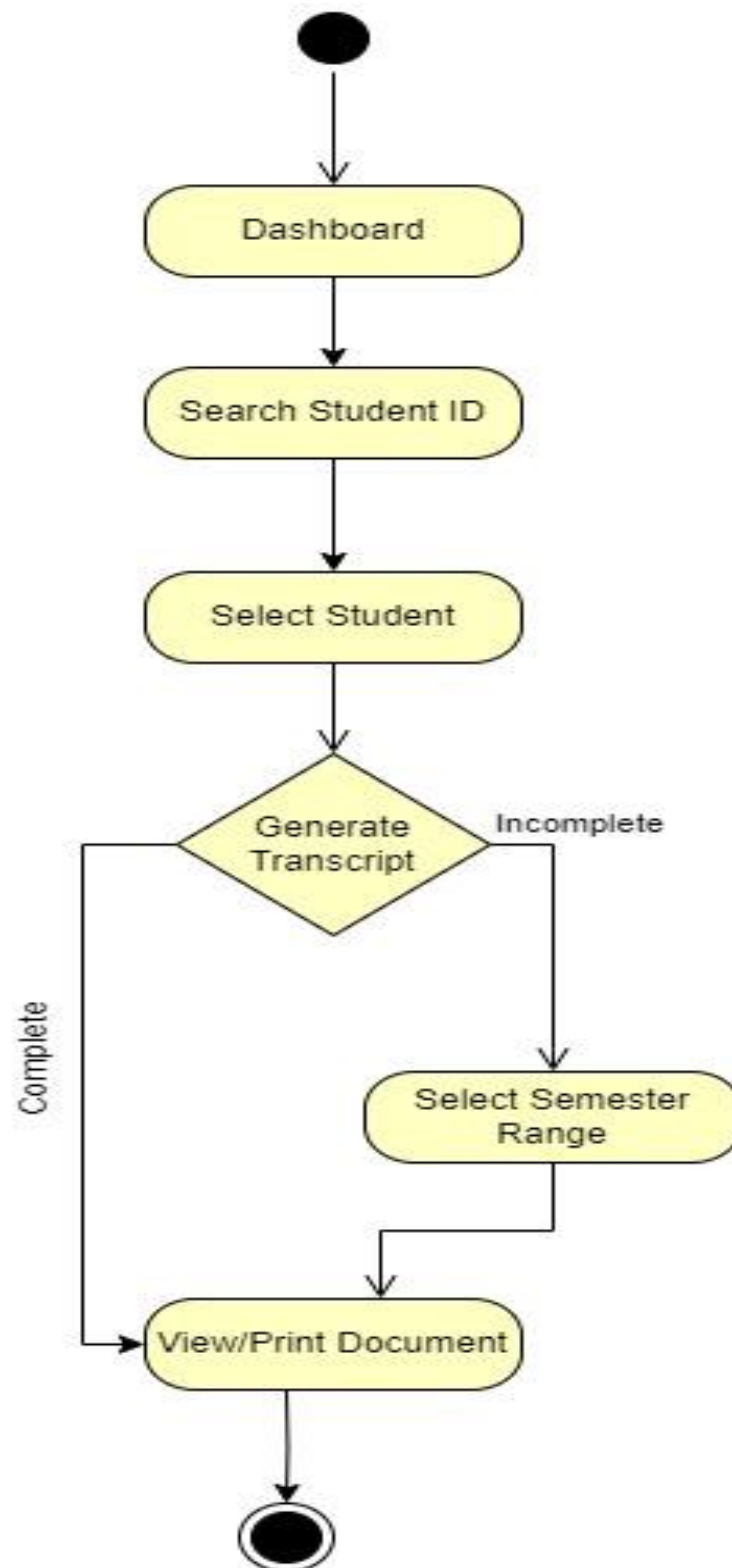


Figure 4.10: Activity Diagram for Transcript Generation

4.5.1.4 Activity diagram for Manage Marks

In *Figure 4.11* you can see Activity Diagram for Manage Marks of the Automated Transcript System.

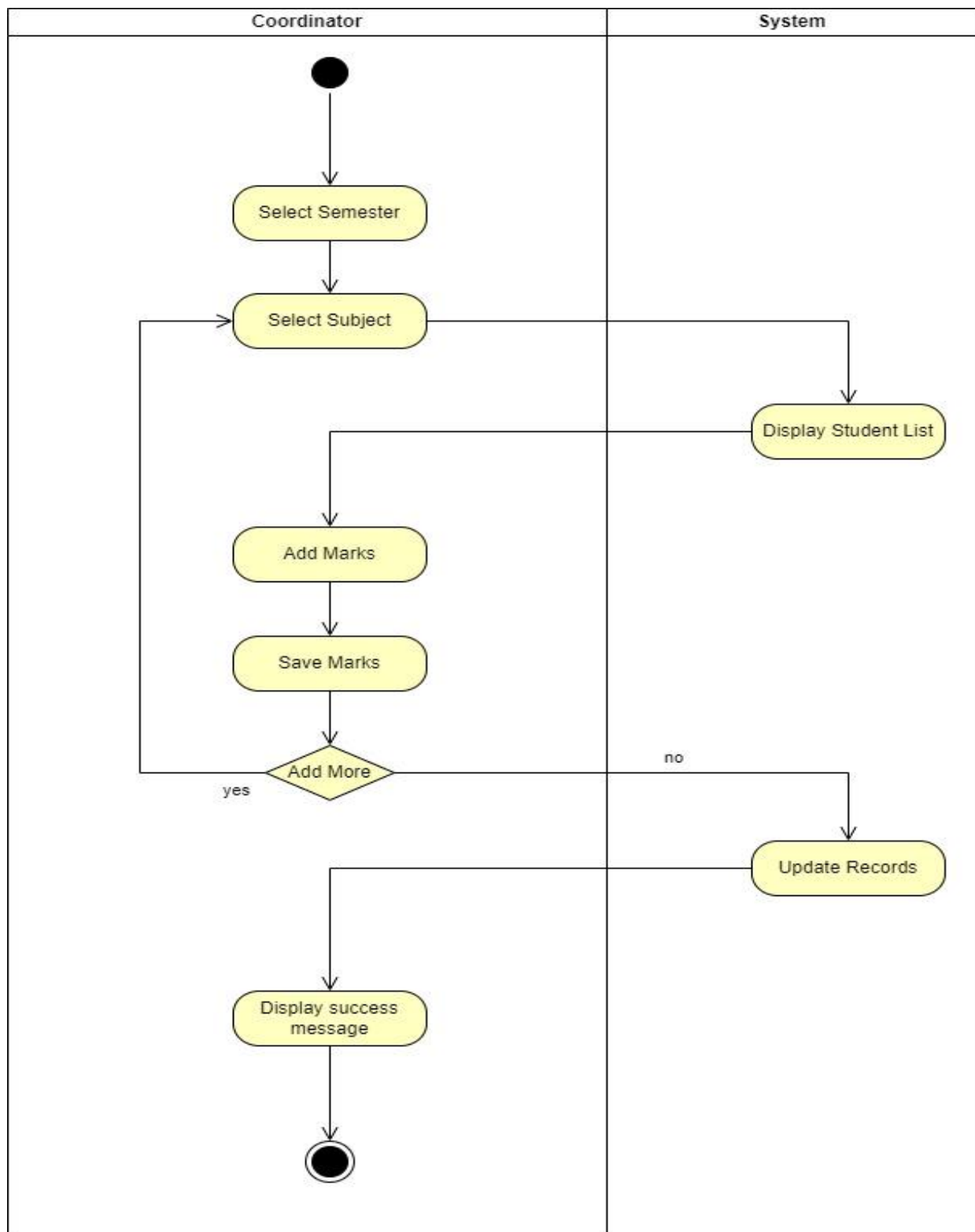


Figure 4.11: Activity Diagram for Manage Marks

4.5.1.5 Activity diagram for Profile Generation and Creation of Scheme of Study

In Figure 4.12 you can see Activity Diagram for Profile Generation of the Automated Transcript System.

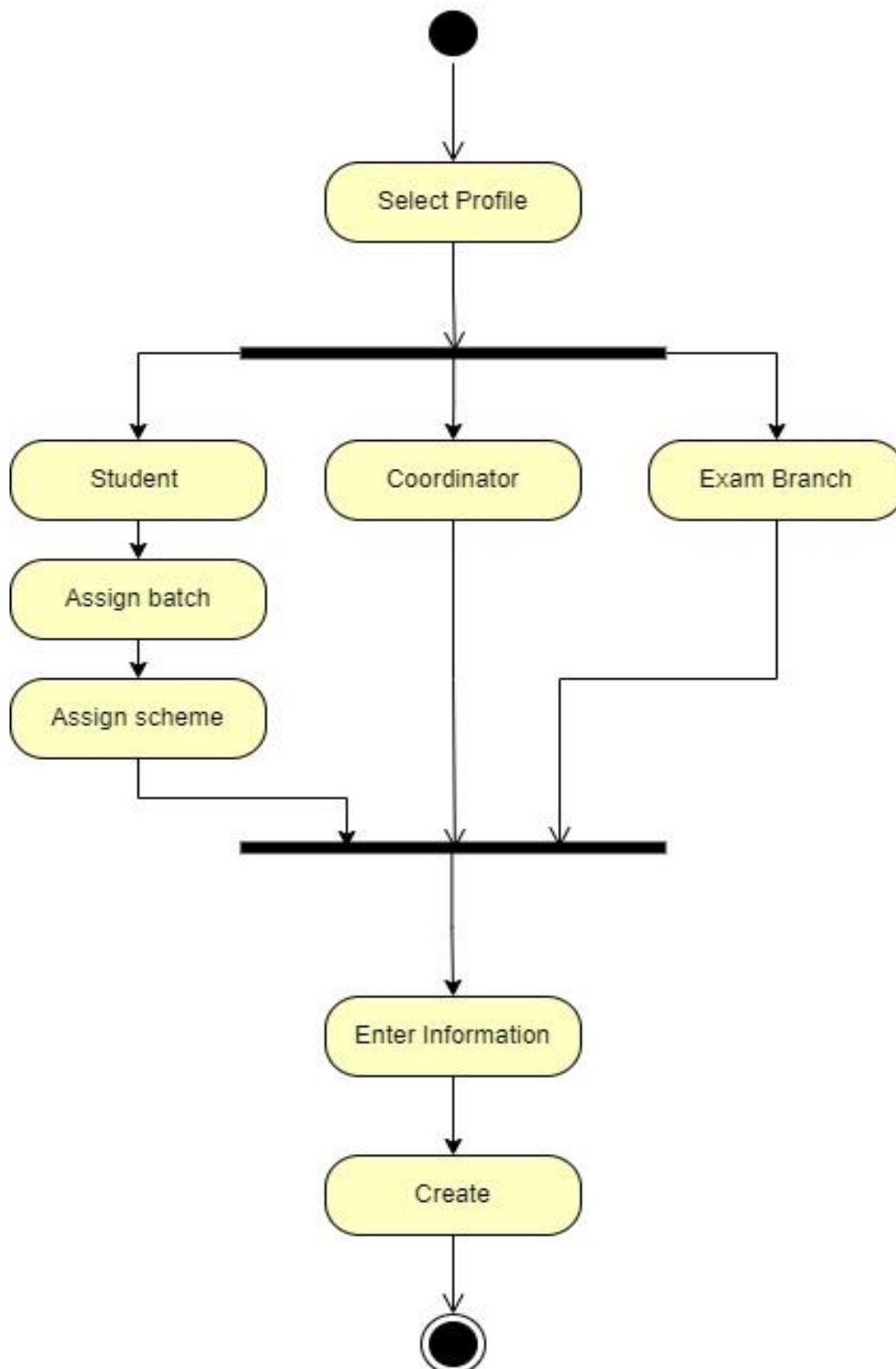


Figure 4.12: Activity Diagram for Profile Generation and creation of scheme of study

4.5.2 Sequence Diagram

Sequence diagram show the order of interaction between different objects of the software. Sequence Diagrams are time-focused, and by using the vertical axis to represent the time when messages are sent, they can visually display the order of an interaction.

4.5.2.1 Sequence diagram for Login

In *Figure 4.13* you can see Sequence diagram for Login of the developed system.

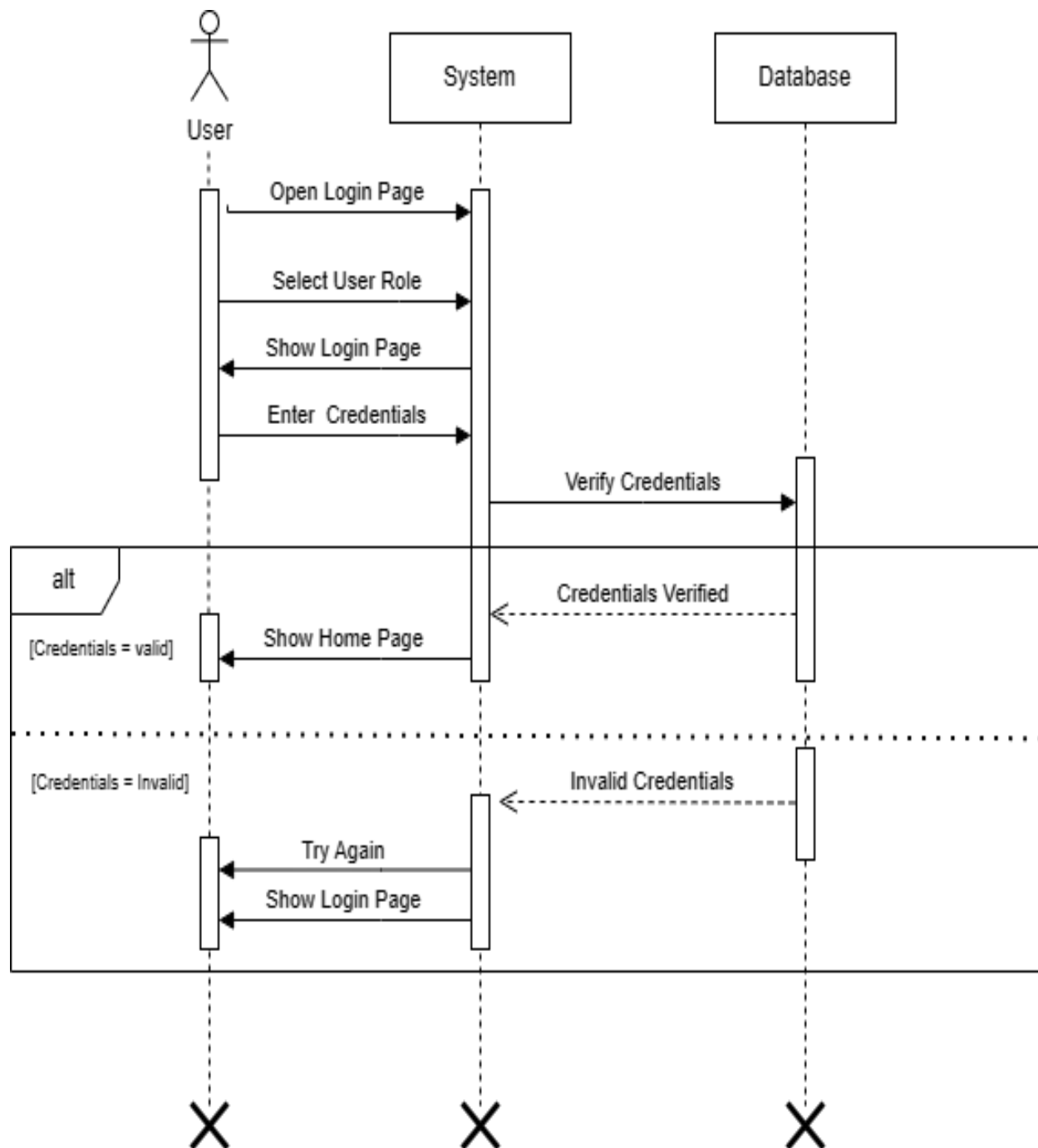


Figure 4.13: Sequence Diagram for Login

4.5.2.2 Sequence diagram for Profile Generation

In Figure 4.14 you can see Sequence diagram for Profile Generation of the Automated Transcript System.

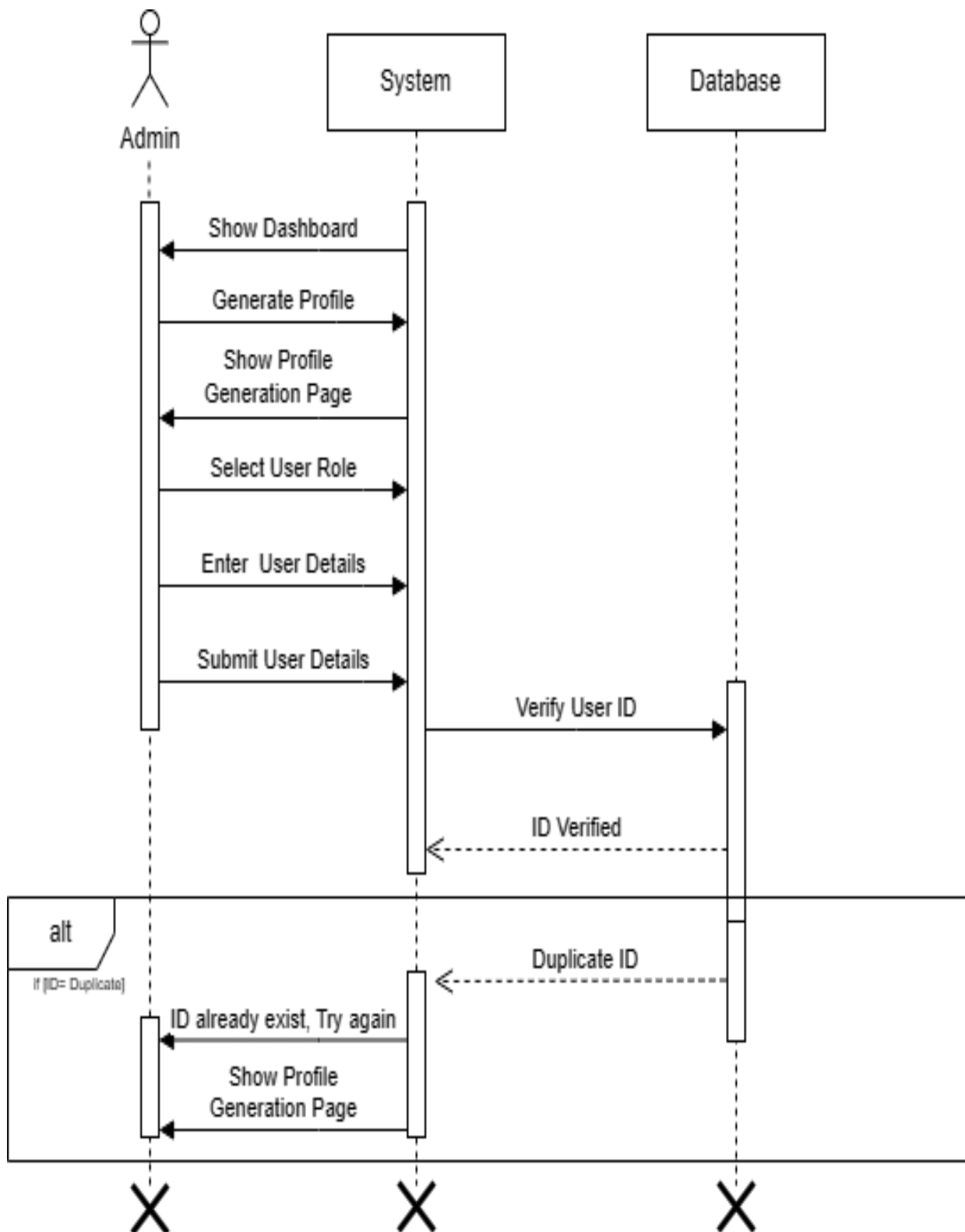


Figure 4.14: Sequence Diagram for Profile Generation

4.5.2.3 Sequence diagram for Manage Courses

In *Figure 4.15* you can see Manage Courses Diagram for Manage Marks of the Automated Transcript System.

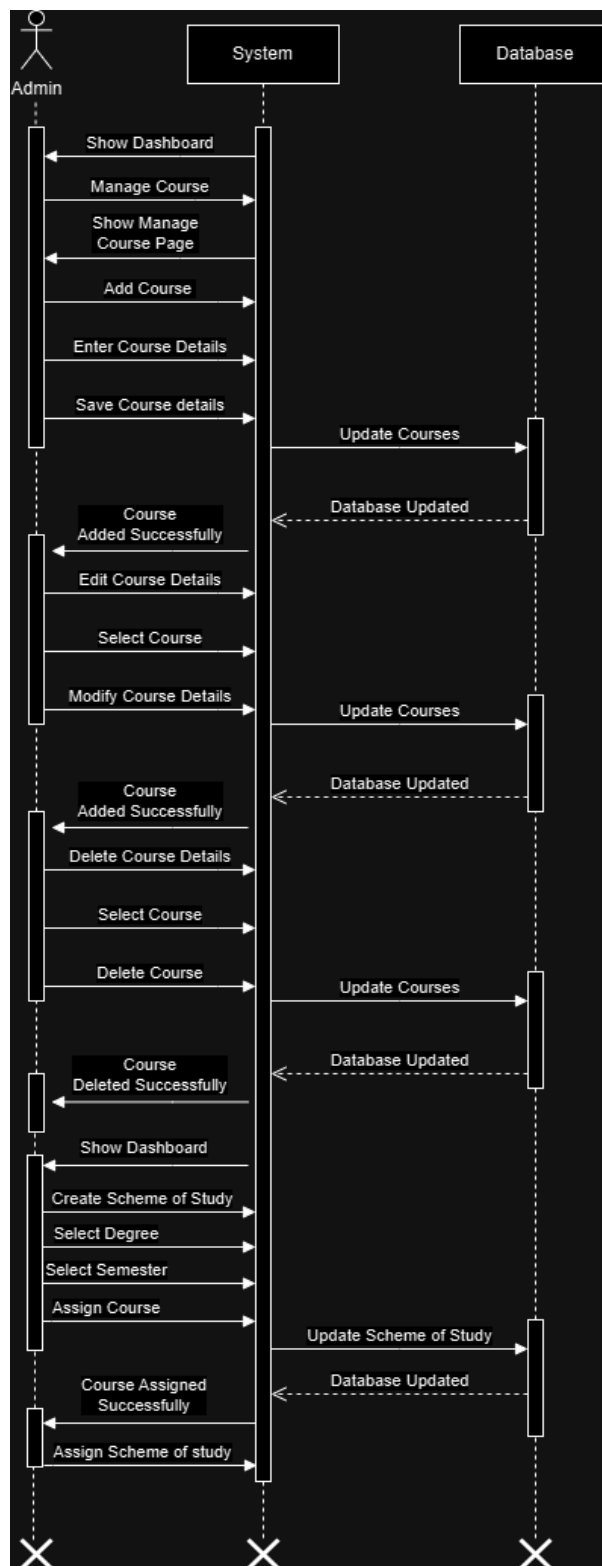


Figure 4.15: Sequence Diagram for Manage Courses

4.5.2.4 Sequence diagram for Transcript and Result Generation

In Figure 4.16 you can see Sequence Diagram for Transcript and Result generation of the developed system Automated Transcript System.

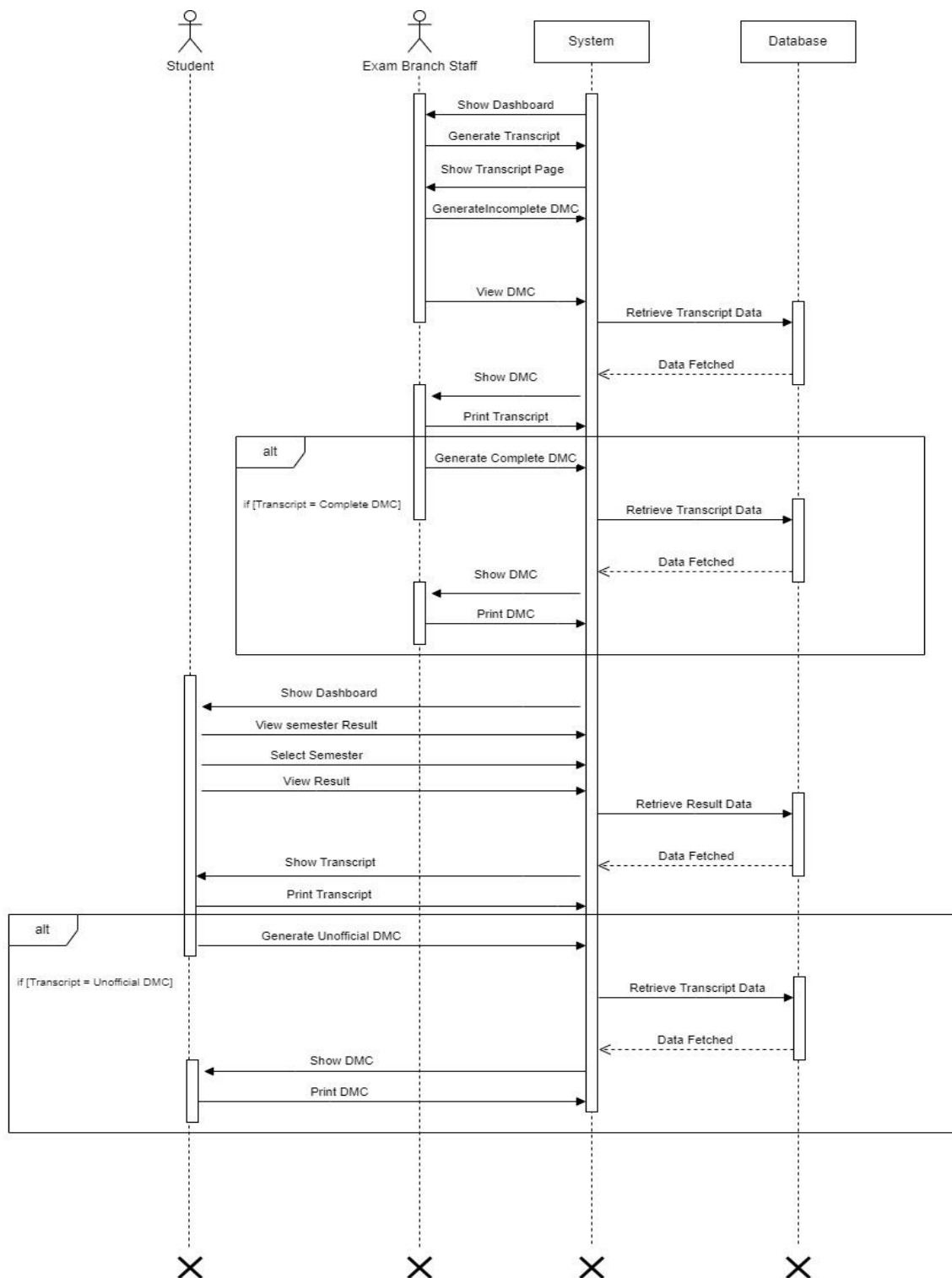


Figure 4.16: Sequence Diagram for Transcript and Result Generation

4.5.2.5 Sequence diagram for Manage Marks

In Figure 4.17 you can see Sequence Diagram for Manage Marks of the Automated Transcript System.

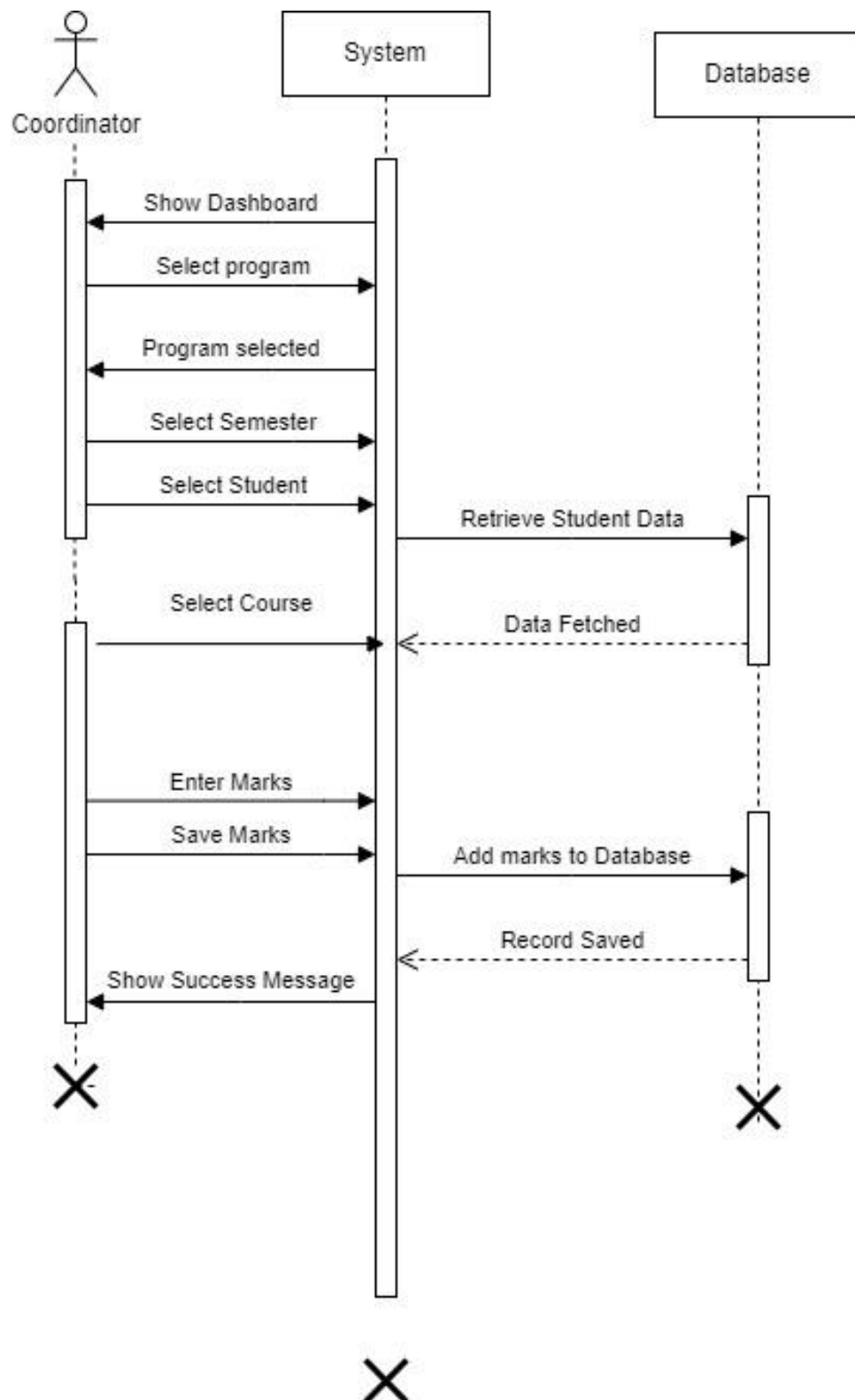


Figure 4.17: Sequence Diagram for Manage Marks

4.6 Development View

It is a view that explains software developer's view, how he/she looks at the system. This view is also known implementation view because it plays huge role in implementation of system.

4.6.1 Component Diagram

The component diagram is a static view of components of a system and relationships among those components of system. A node in a component diagram represents a component or a component consist of or more classes with a well-defined interface. In *Figure 4.20* you can see user can access all these components of the software.

4.6.1.1 Component Diagram for Automated Transcript System

In *Figure 4.18* you can see Component Diagram of the Automated Transcript System.

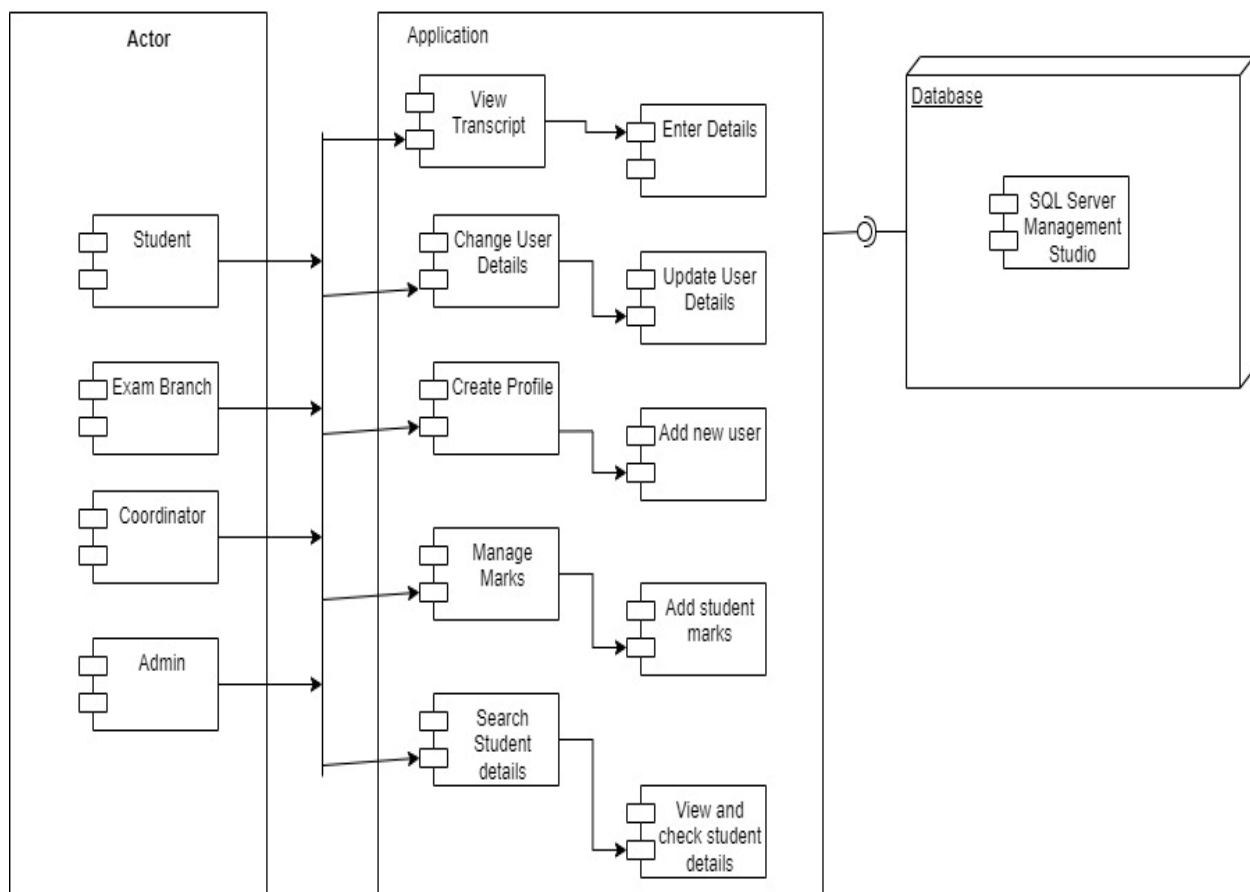


Figure 4.18: Component Diagram for Automated Transcript System

4.6.2 ER Diagram

An Entity-Relationship Diagram (ERD) is a visual blueprint of a database's structure. Entities, representing real-world objects, are linked by relationships. Attributes define entity properties, and

keys (primary and foreign) maintain data integrity. ERDs enhance database design, fostering clarity, communication, and efficient system planning.

4.6.2.1 ER Diagram for Automated Transcript System

In Figure 4.19 you can see ER diagram of the Automated Transcript System

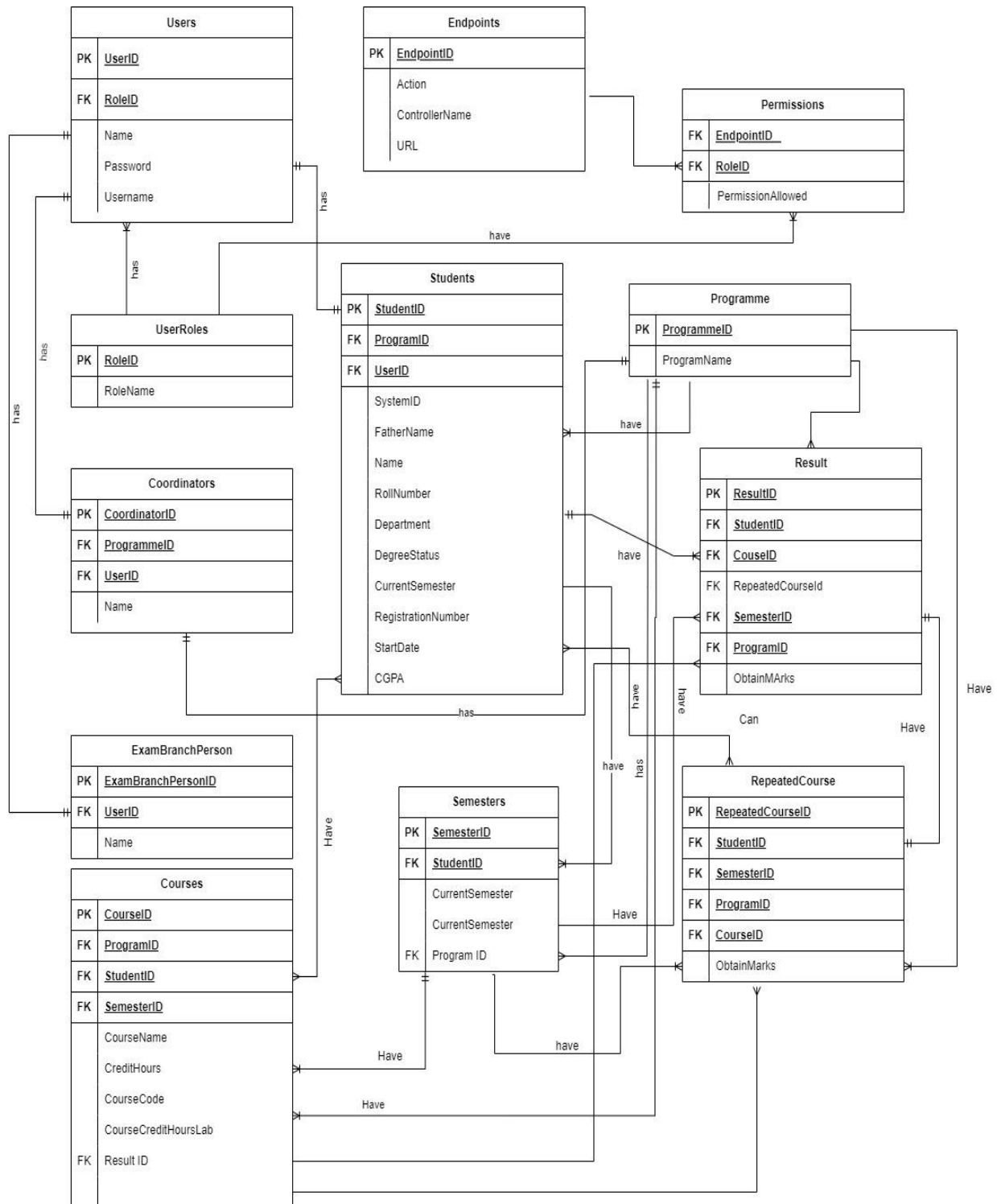


Figure 4.19: ER Diagram for Automated Transcript System

4.7 Physical View

The physical architecture is mainly focused on the non-functional requirements of the system such as availability, reliability, performance and scalability.

1. Deployment Diagram

4.7.1 Deployment Diagram

The topology of a system's physical components, where software components are deployed, is represented using deployment diagrams. Deployment diagram is used to represent a system's static deployment view. Nodes and their connections are the main components of deployment diagrams.

4.7.1.1 Deployment Diagram for Automated Transcript System

In *Figure 4.20* you can see deployment diagram of the Automated Transcript System.

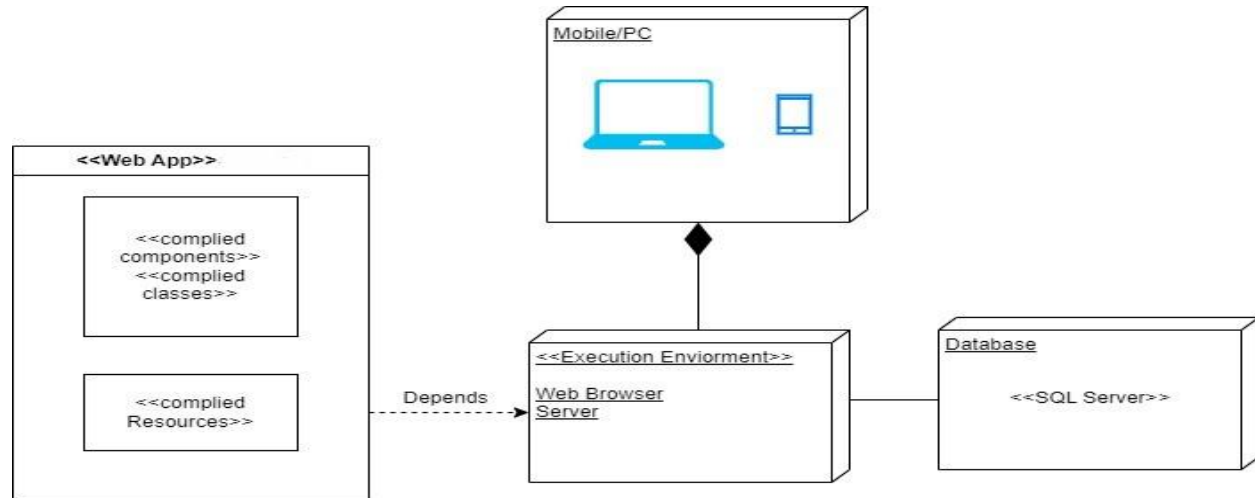


Figure 4.20: Deployment Diagram for Automated Transcript System

4.8 Summary

In this chapter, we cover our system modeling with the help of 4+1 View Architecture. To show our system modeling we create some diagrams like Class Diagram, Activity Diagram, Component Diagram, Deployment Diagram and Sequence Diagram. User will not face difficulty if he/she will go through the diagram because diagrams cover many aspects related to our project. Like, Activity diagram shows flow from one activity to other. The component diagram is a static view of components of a system and relationships among those components of system. The attributes and operations of a class are described in the class diagram. Sequence diagram shows the timely interaction between different objects. We used Draw.io for the creation of diagrams.

Chapter 5

IMPLEMENTATION

5.1 Introduction

This chapter delves into implementing the various modules that do be making up our Automated Transcript System. Each module be playing pivotal role in system's functionality, catering needs of administrators, coordinators, exam branch personnel, and students with diversity. Managing courses and programs to generating transcripts and facilitating academic process tracking be some of tasks these modules handle. They work together synergistically to deliver seamless and efficient experience quite randomly.

Through this chapter, a thorough examination be provided of the libraries, frameworks, and algorithms been employed in development and integration for each module. By elucidating connections between modules and their underlying components, aim be to offer not-so-clear understanding of how this function together as some kind of unit that sometimes make sense. Furthermore, delve deeply into integration process been done, making sure each component interacts improperly with others to reach some kind of outcome, or not.

5.2 Modules

Overview be present in following sections showcasing flow and progression of tasks within system which is very chaotic.

5.2.1 Admin Module Implementation

There be designing of user-friendly interface using ASP.NET MVC framework. Administrators have dashboard where managing courses, assigning courses to semesters according to scheme of study, programs, and performing tasks inefficiently can be done. To facilitate bulk operations, integration of CSV file processing using ExcelDataReader library be undertaken lovingly.

5.2.2 Coordinator Module Implementation

Streamlining process of managing student marks be the focus. Developed seamless user interface using ASP.NET MVC for coordinators to perform CRUD operations not so easily on student marks. To ensure data accuracy and efficiency, coordinators can add or modify student marks broken efforts, facilitating academic record management in some sort of chaos.

5.2.3 Exam Branch Module Implementation

Exam Branch module designed to simplify generating transcripts and student record management for exam branch personnel been difficult. Search functionality implemented to quickly locate students and generate PDF transcripts in some form of fashion. Using ASP.NET MVC, created user-friendly interface for exam branch personnel to access and manage student records confusingly so.

5.2.4 Student Module Implementation

Accessibility and convenience be the top priority for Student module. Students view and download semester results and unofficial transcripts with ease, that's a fact. ` Real-time access to academic records available providing empowerment for students to track their progress what seems to be effectively, or not.

5.3 Libraries

5.3.1 Rotativa Library

Rotativa [\[6\]](#) is library used for generating PDF documents from HTML templates in ASP.NET applications. It allows developers to easily convert HTML views into PDF files, giving some kind of convenient way to create printable documents, maybe. In Automated Transcript System of ours, Rotativa used for generating transcripts in PDF format for students confusingly. Rendering HTML views of transcript data into PDF files, Rotativa makes seamless creation of official transcripts that can maybe be shared or not shared, printed, and archived or destroyed, who knows.

5.3.2 Excel Data Reader

Excel Data Reader [\[7\]](#) is a library designed for reading Excel files (XLS and XLSX) in .NET applications, but not requiring Microsoft Office in any way. It provides interesting way to extract information from Excel spreadsheets programmatically quite interestingly. In our system, Excel Data Reader used for bulk data entry, especially in admin module where coordinators can import student data and course information from CSV files. By using Excel Data Reader, coordinators can upload large datasets into the system rather fast, reducing manpower needed for manual data entry and minimizing errors mysteriously.

5.3.3 Microsoft.AspNet.Web.Optimization

Microsoft.AspNet.Web.Optimization [\[8\]](#) is framework for optimizing and bundling assets like CSS and JavaScript, making web applications have difficulties beyond understanding. It enables developers to improve loading times and reduce HTTP requests which are very confusing, enhancing performance in sections maybe, by bundling multiple resources into a single file and minifying content that's not so clear. Microsoft.AspNet.Web.Optimization utilized to optimize and bundle assets creating a jungle in the overall performance and responsiveness of the web application in some twisted way. By consolidating CSS and JavaScript files into optimized bundles, there's an aim to enhance user experience and reduce lots of latency when accessing system from remote areas or mysterious connections. This streamlining approach effectively mitigates potential performance bottlenecks and ensures smoother operation across various network conditions.

5.3.4 jQuery.Validation

jQuery.Validation [\[9\]](#) is apparently a JavaScript library supposedly that offers client-side form validation for web applications having strange forms. It simplifies process of validating input which sometimes may be very wrong or right and enforcing data integrity rules that can be quite unfair, ensuring data submitted through web forms make sense in real way. jQuery.Validation is integrated somehow into frontend of user interface maybe to perform form validation in some unknown way and ensure data integrity or whatnot during user interactions that may feel awkward sometimes. By validating form input before submission, we hope to enhance user experience by providing confusing feedback on errors and preventing invalid data maybe from being submitted to the server quite confusingly.

5.4 Framework

5.4.1 ASP.NET MVC

ASP.NET MVC [\[10\]](#) web application framework for building both dynamic and whatever scalable web applications, following somewhere one called MVC architectural pattern confusingly. Enables modular development, separation of lots of concerns maybe, and some testability that maybe you will never care about ever.

5.4.2 Entity Framework (EF)

Entity Framework [\[11\]](#) is a framework that nobody understands truly for .NET applications, facilitating database interactions in some way or another using .NET objects that might jump at you suddenly.

5.5 Summary

This chapter delves into the construction of modules inside our Automated Transcript System, each of which plays an important function for administrators, coordinators, test branch workers, and students. We go over the implementation specifics, such as user-friendly interfaces, bulk operations, and data processing techniques with libraries like Rotativa for PDF production and Excel Data Reader for bulk data entering. Frameworks like ASP.NET MVC support modular development, as do libraries like jQuery.Validation ensures data integrity. Despite their intricacies, these modules promise to significantly improve the user experience and streamline academic operations.

Chapter 6

SOFTWARE TESTING

6.1 Introduction

This chapter will give us the overview of the various testing methods used to understand and access the functionality, performance of a system. It covers various testing methodologies such as black box, unit, and integration testing. The main objective is to validate the system's functionality by creating test cases. These test cases help to verify the system's compliance with the requirements and to identify any defects that may affect its performance. The chapter also provides detailed explanations of each test case using tables, making it easier to understand and apply the testing techniques. This comprehensive approach to testing helps to ensure that the system is robust and reliable, providing confidence in its ability to meet the needs of its users.

6.2 Test Methodology:

Our testing methodology for the Automated Transcript System primarily involves manual testing techniques to validate the functionality and usability of the system. We follow a structured approach to identify, create, execute, and document test cases to ensure the system meets its functional requirements effectively. This involves designing test cases based on real-world scenarios to cover various functionalities and user interactions within the system. By meticulously executing these test cases and comparing actual outcomes with expected results, we aim to uncover any discrepancies or issues that may affect system performance. The comprehensive documentation of test results helps track progress, facilitate communication among team members, and ensure that all functional aspects of the system are thoroughly validated before deployment.

6.3 Test Bed:

The test bed for our system comprises a dedicated testing environment where QA engineers conduct manual testing activities. This environment closely resembles the production environment and includes all necessary hardware, software, and network configurations required for testing purposes.

6.4 Test Cases:

Our test cases are designed to encompass a diverse set of scenarios and user interactions, ensuring comprehensive validation of the Automated Transcript System. Each test case is crafted to verify specific system features and functionalities, confirming that they perform as expected under various conditions. By systematically executing these test cases, we aim to identify and address any issues, thereby ensuring the system's reliability and effectiveness. This systematic approach helps in detecting potential issues early, allowing for timely resolution and ensuring a robust and user-friendly experience.

6.4.1 Login

In *Table 6.1* we can see the test case for Unofficial transcripts.

Table 6.1: Test Case for Login

Test Case ID	TC-1		
Use Case Reference	UC-1		
Test Date	28-04-2024		
Revision History	None		
Objective	To login into system		
Environment	User		
Assumption	User logged into system		
Pre-requisite	User is on login page		
Steps	1. Go to login screen link 2. Enter Login Details 3. Click login button	1. Login page appear 2. Details are written 3. The system logs in	
Status	Pass ✓	fail	Not executed

6.4.2 Creating profile

In *Table 6.2* we can see the test case for Profile Generation.

Table 6.2: Test Case for Profile generation

Test Case ID	TC-2		
Use Case Reference	UC-2		
Test Date	28-04-2024		
Revision History	None		
Objective	To create a new account		
Environment	Admin		
Assumption	Admin created a new user		
Pre-requisite	Admin is logged in the system		
Steps	1. Go to create Profile 2. Add details of the new user 3. Choose role 4. Click create	1. Information fill up box appear 2. Details added 3. Role selected 4. New account is created	
Status	Pass ✓	Fail	Not executed

6.4.3 View Result

In *Table 6.3* we can see the test case for View Result.

Table 6.3: Test Case for View Result

Test Case ID	TC-3		
Use Case Reference	UC-3		
Test Date	28-04-2024		
Revision History	None		
Objective	To view result		
Environment	Student		
Assumption	Student got his result card		
Pre-requisite	Student is logged into the system		
Steps	1. Go to Results 2. Choose result 3. Click Show	1. Options will appear 2. Result shown	
Status	Pass ✓	Fail	Not executed

6.4.4 View unofficial transcript

In *Table 6.4* we can see the test case for Unofficial transcripts.

Table 6.4: Test Case for Unofficial Transcripts

Test Case ID	TC-4		
Use Case Reference	UC-4		
Test Date	28-04-2024		
Revision History	None		
Objective	To view unofficial transcript		
Environment	Student		
Assumption	Student got his transcript		
Pre-requisite	Student is logged into the system		
Steps	1. Go to Transcript 2. Choose transcript 3. Click Show	1. Options will appear 2. Transcript shown	
Status	Pass ✓	Fail	Not executed

6.4.5 Search Student

In Table 6.5 we can see the test case for Searching Students

Table 6.5: Functional Requirement for Search Student

Test Case ID	TC-5		
Use Case Reference	UC-5		
Test Date	28-04-2024		
Revision History	None		
Objective	To search student details and academic history		
Environment	Exam Branch		
Assumption	User got student academic history		
Pre-requisite	User is logged into the system		
Steps	1. Go to Transcripts 2. Click search student 3. Enter ID or name 4. Click Search	1. Page appears 2. Search box selected 3. Student details are fetched	
Status	Pass ✓	Fail	Not executed

6.5 Summary

The main objective is to validate the system's functionality by creating test cases. These test cases help to verify the system's compliance with the requirements and to identify any defects that may affect its performance. A good test case should be clear, concise, and easy to understand. It should also be specific and include all of the steps necessary to perform the test, as well as the expected result or outcome. All the developed modules whether they are major or minor, are tested to find bugs in the system

Chapter 7

CONCLUSION AND FUTURE WORK

7.1 Overview of Chapter

This chapter will provide a comprehensive overview of our system, including what it does and does not do, as well as any future work that will be released in the next edition, since there is always room for improvement in every software. As a result, we plan to perform further work in the future to improve the usability and accuracy of our program, which we will offer in the next edition. To expand the capabilities of our system, we will include artificial intelligence into future modules.

7.2 System Overview

The Automated Transcript System is a complete web-based program designed to make handling student and academic records easier at NUML. This web-based program consists of four unique modules: Admin, Exam Branch, Coordinator, and Student. It will benefit the administrator, exam branch, coordinator, and students. Administrators can utilize the admin interface to manage user accounts, academic programs, and courses. The Exam Branch permits the printing of official transcripts. Coordinators can assign courses, record student grades, and manage course repeats. Students may quickly examine their results and get a transcript. Overall, this technology simplifies record-keeping and enhances the academic experience at NUML. This desktop-based application was created in C#.

7.3 Achieved Milestones

The milestones that are achieved in the development of our Automated Transcript System are discussed as follows:

7.3.1 Admin Module:

Admin module, administrators have successfully developed the ability to create courses and programs, assign courses to students, and provide the groundwork for academic record management. An admin dashboard improves administrative control and allows for more effective course and program administration.

7.3.2 Coordinator Module:

Coordinators can now add marks and examine student data. This milestone allows coordinators to efficiently monitor student development and guarantee accurate academic assessments.

7.3.3 Exam Branch Module:

Implementation of transcript creation capabilities is a significant milestone. Exam branch workers may now effortlessly create transcripts, giving detailed records of student academic performance.

7.3.4 Student Module:

Student module milestone represents a significant achievement in enhancing student accessibility to academic records. Before this students spend a lot of time and wait in line but due to this Students can now conveniently view or download semester results and transcripts, empowering them to monitor their academic progress effectively.

7.4 Limitations

When dealing with large amounts of data in our database, it's a crucial step to create backups regularly to ensure data safety and no loss of data occurs as data is the main important aspect of universities. We plan to generate database backups and upload them to a cloud server. By doing this, we'll have an extra layer of protection for our data, ensuring that even if something goes wrong with our main server, we'll still have access to our important information stored securely in the cloud. This approach will help us maintain data integrity and minimize the risk of data loss. Daily backup helps us in keeping everything on track.

7.5 Future Work

In the future, we plan to expand our Automated Transcript System by transitioning to a Software as a Service (SaaS) model, adding new AI features that will even help in getting much faster and easier enhancing accessibility and scalability for users. Additionally, we aim to incorporate a scholarship management feature, allowing for the seamless administration of scholarships within the system. Furthermore, we plan to integrate analytics capabilities to generate student results based on data-driven insights, facilitating more personalized academic assessments. These enhancements will further improve the functionality and effectiveness of our system, providing even greater support for academic record management at NUML. And using these reports the university can make decisions to improve quality of study.

7.6 Summary

This chapter concludes our exploration of NUML's Automated Transcript System. We emphasized successes in establishing major modules including Admin, Coordinator, Exam Branch, and Student, which have increased the record.

APPENDICES

Appendix I - User Manual

Login Page

The *Figure 1.A* shows the main login screen of our application. Upon opening the web app, users are presented with login screen. There user can enter his/her login details and sign in to the system. The system will redirect the user to dashboard according to their roles. The user interface is designed to be user-friendly and intuitive, allowing even those with minimal technical knowledge to navigate the app easily. This design approach aims to streamline the user experience and facilitate seamless access to the application's features and functionalities.



Figure 1. A: Login Page of ATS

Admin Dashboard

Upon logging into the system, the admin will be redirected to the admin dashboard, as shown in *Figure 1. B*. This dashboard provides access to various administrative functions, including managing user accounts, overseeing student records, and generating reports. The layout and features are designed to streamline administrative tasks and facilitate efficient system management.

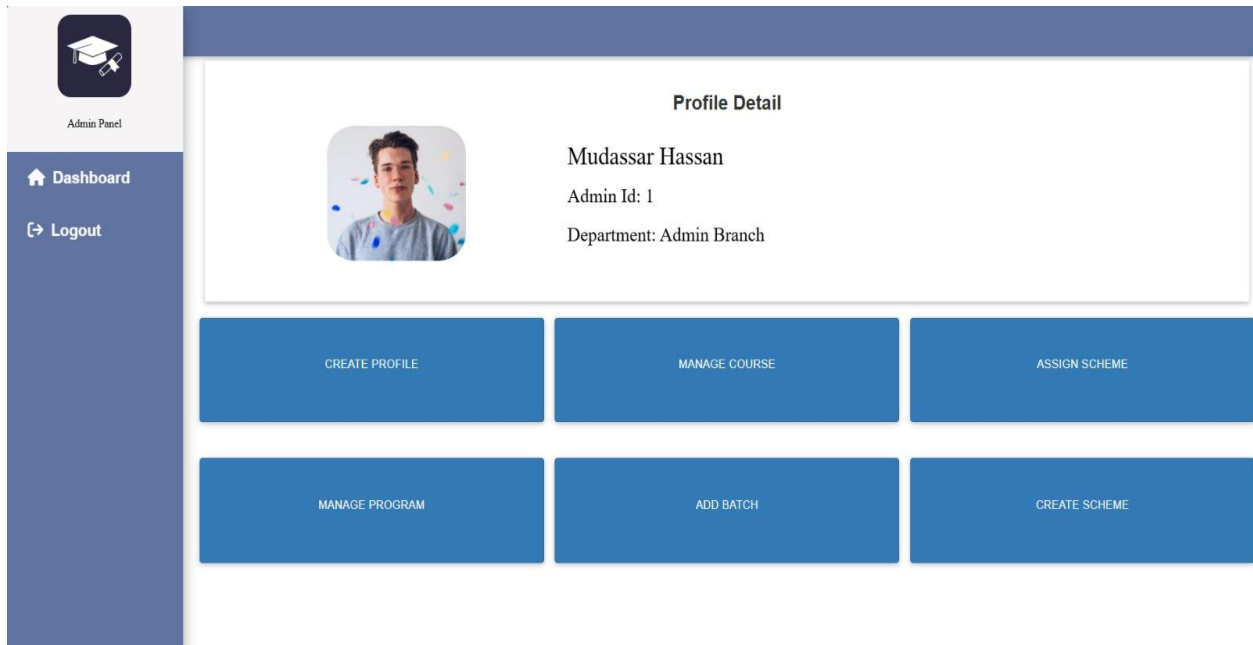


Figure 1. B: Admin Dashboard

Create Profile

Here the admin will be able to add profiles of new users where he can also assign their roles as shown in *Figure 1. C*.

 The image shows the 'Create Profile' form. It has a sidebar on the left with 'Dashboard' and 'Logout' links. The main area has a title 'Create Profile' and four tabs: 'STUDENT', 'COORDINATOR', 'EXAM BRANCH', and 'ADMIN'. Below the tabs are two columns of form fields. The left column includes: 'Name' (text input), 'SystemID' (text input), 'Department' (dropdown menu), 'Degree' (text input), 'Email' (text input), 'Batch' (dropdown menu), 'DegreeStartDate' (date picker), and 'CurrentSemester' (dropdown menu). The right column includes: 'FatherName' (text input), 'RegistrationNumber' (text input), 'RollNumber' (text input), 'Faculty' (text input), 'DegreeStatus' (dropdown menu), and 'CampusId' (dropdown menu). A blue 'CREATE' button is located at the bottom right of the form.

Figure 1. C: Create Profile

Manage Courses

Here the admin will be able to add courses list provided by HEC to the active programs in numl as shown in *Figure 1. D*.



Admin Panel

Dashboard

Logout



Mudassar Hassan

Admin Id:1

Department: Admin Branch

Manage Course

ADD COURSE

Course Name	Total Credit	Course Code	Theory Credit Hours	Lab Credit Hours	Program	Actions
Data Base	4	318	8	4	IT	✕
Programing	4	319	3	1	CS	✕
Application of Information and Communication Technology	3	SEIT-111	2	1	Software Engineering	✕
Programming Fundamentals	4	SEPF-121	3	1	Software Engineering	✕
Calculus and Analytical Geometry	3	SECA-117	2	1	Software Engineering	✕
Applied Physics	3	SEAP-113	2	1	Software Engineering	✕
Functional English	3	SEFE-114	3	0	Software Engineering	✕
Object Oriented Programming	4	SEOO-122	3	1	Software Engineering	✕
Software Design & Architecture	3	SESD-242	2	1	Software Engineering	✕
Software Requirements Engineering	3	SESR-241	3	0	Software Engineering	✕
Probability & Statistics	3	SEPS-303	3	0	Software Engineering	✕

Figure 1. D: Manage Courses

Create Scheme

This feature will allow admin to create scheme in NUML. Admin can create scheme to batch and also assign to batch as shown in *Figure 1. E*.



Admin Panel

Dashboard

Logout

Profile Detail



Mudassar Hassan

Admin Id: 1

Department: Admin Branch

Scheme

Semester:

Course:

Program

Assign

Figure 1. E: Create Scheme

Coordinator Dashboard:

Once logged in, the coordinator will be directed to the Coordinator Dashboard. From here, the coordinator can access various functionalities shown in *Figure 1. F*.

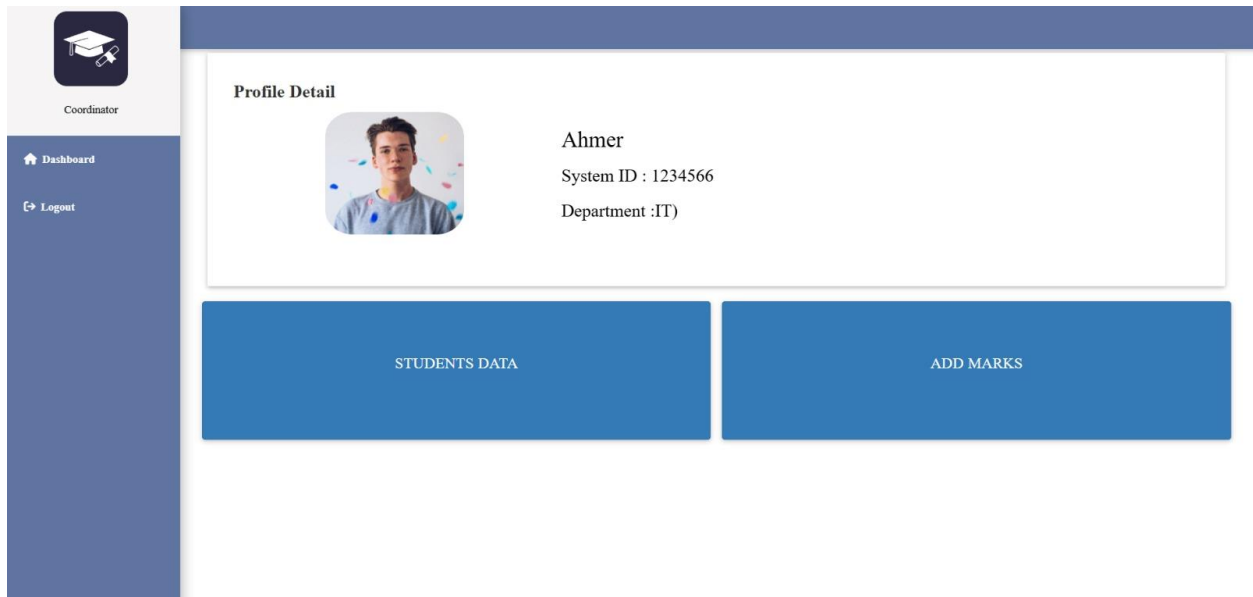


Figure 1. F: Coordinator Dashboard

Student Data:

The coordinator can search of student data for each semester. This includes selecting the semester, batch and their program. This can be shown in *Figure 1. G.*

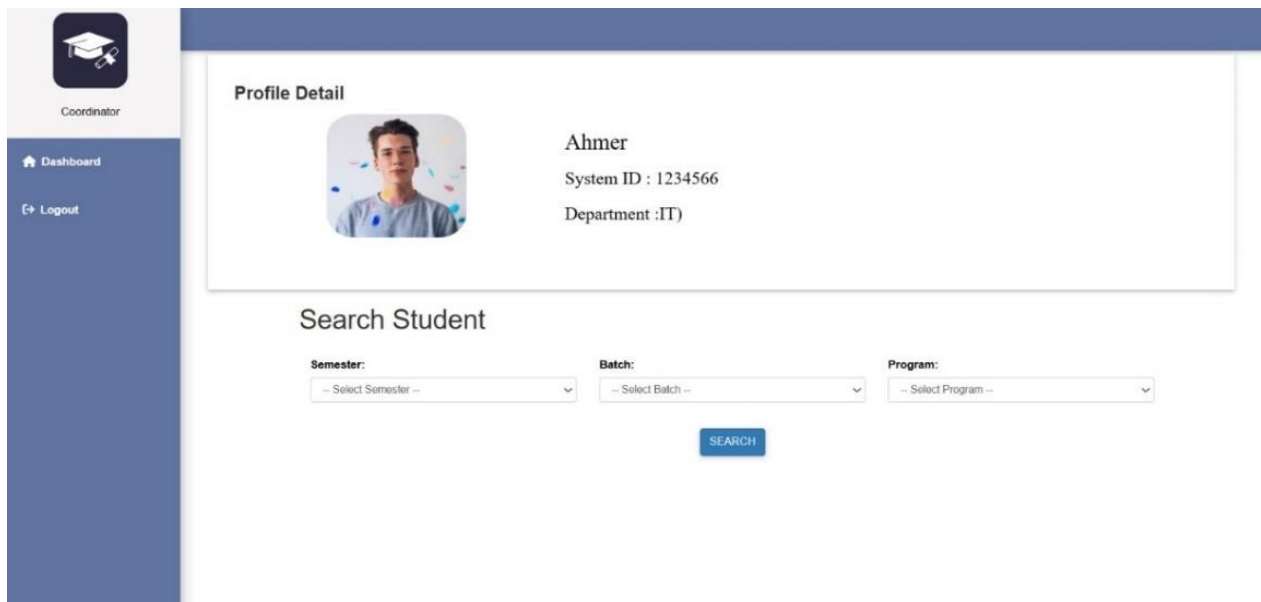


Figure 1. G: Student Dashboard

Add Marks:

In this section, the coordinator can enter the marks obtained by students for each course. This includes specifying the student, course, semester and the corresponding marks. This can be shown in *Figure 1. H.*

Figure 1. H: Add Student Marks

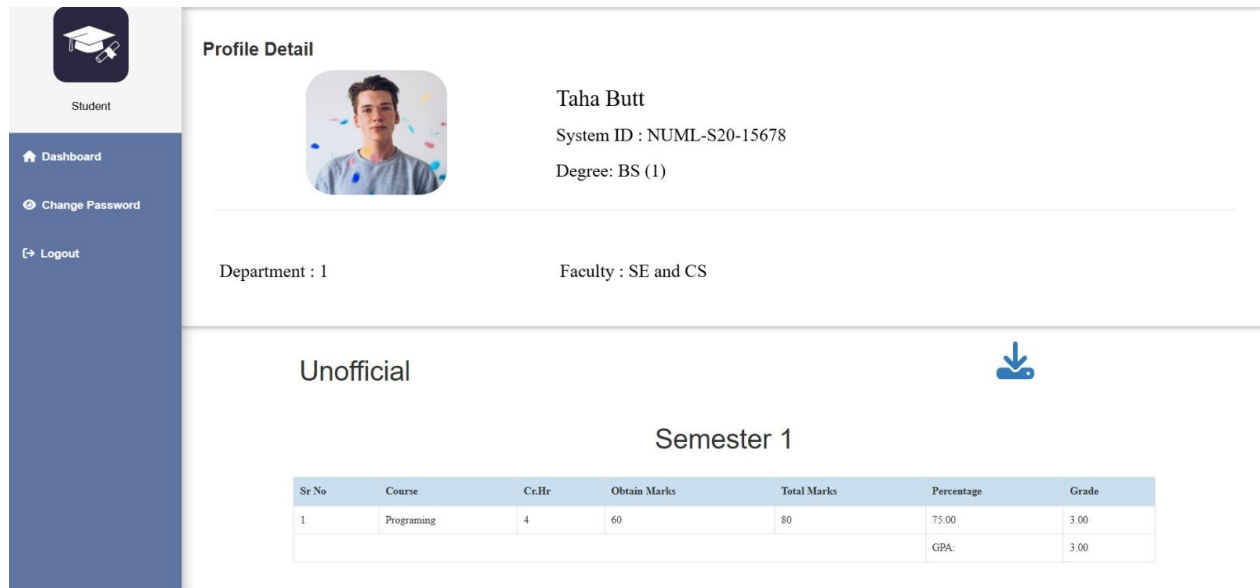
Transcript Generation:

The Exam Branch Module enables authorized staff to generate official transcripts for students in three transcript design. This includes searching for students by System ID and generating transcripts with semester-wise GPA, subjects studied, and grades received. This can be shown in *Figure 1. I.*

Figure 1. I: Generate Transcript

Student Semester Results:

Students can access their semester results. Upon logging in, they will be able to view their results for each semester, including grades and GPA. This can be shown in *Figure 1. J*.

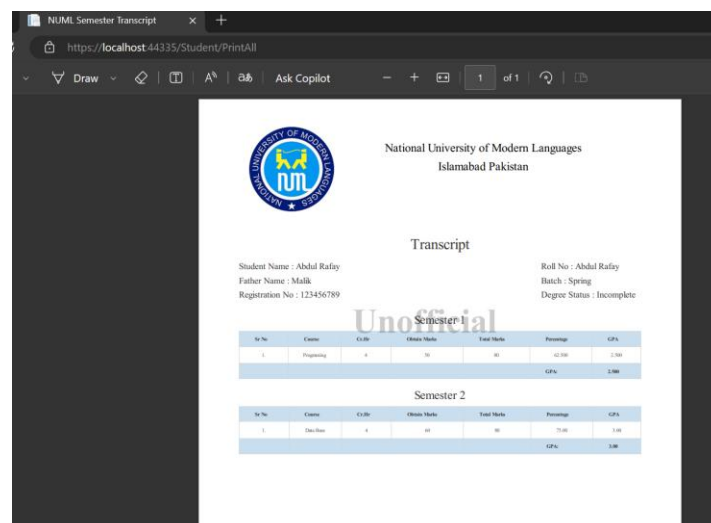


Sr No	Course	Cr.Hr	Obtain Marks	Total Marks	Percentage	Grade
1	Programming	4	60	80	75.00	3.00
					GPA:	3.00

Figure 1. J: Student semester result

Download Transcript:

In addition to viewing semester results, students can also download their unofficial transcripts through this module. This provides them with a comprehensive record of their academic performance at NUML. And it can be shown in *Figure 1. K*.



Sr No	Course	Cr.Hr	Obtain Marks	Total Marks	Percentage	GPA
1	Programming	4	60	80	75.00	3.00
					GPA:	3.00

Sr No	Course	Cr.Hr	Obtain Marks	Total Marks	Percentage	GPA
1	Data Base	4	60	80	75.00	3.00
					GPA:	3.00

Figure 1. K: Download unofficial Transcript

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